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The κ -Chain: A Sovereign Islamic Financial Infrastructure Layer

White Paper v0.7.8 — June 2026

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v0.7.8 corrections (2026-06-15): Four substrate claims hardened against the companion simulation framework (Paper 3 §“Systemic and adversarial robustness,” four cadCAD models; reproducible from the seeded scripts in simulations/). (i) §11.2 the κ thin-market fallback redesigned from a hard volume-threshold switch to a continuous gated blend — a hard switch is whipsawable across the threshold (a Monte-Carlo liquidity-whipsaw attack jumps κ 165 bp in a single block and trips the margin-call line $13\times$ per run); the continuous handover ($\kappa = w \cdot \kappa_{\text{perp}} + (1-w) \cdot \kappa_{\text{sukuk}}$, $w = \text{smoothstep}$ on an asymmetric-EMA volume weight + a validator-dispersion gate) holds the worst single-block move below the liquidation trigger with zero cliffs. (ii) §5.3 $\times/\text{oracle}$ median robustness now simulation-validated (stake-weighted median holds κ to 2–4 bp at $f \leq 1/3$; a withholding attack past $1/3$ yields a safe pause, never a corrupted κ). (iii) New §11.2 cross-market contagion risk + mitigation (a shared-collateral cascade across κ -markets is threshold-like; the ± 75 bp clamp + liquidation throttle + takaful collateral floor keep a one-market shock sub-critical). (iv) The κ -Compute *salam* backstop sizing (8% tabarru‘) noted as a forward extension of the same takaful machinery.

v0.7.7 corrections (2026-06-14): Matching-engine design finalised after an adversarial design review (dYdX v4 off-chain memclob vs Injective on-chain FBA, benchmark-design first principles, cross-checked against Papers 1/3/13 and the audited contracts). §5.2 $\times/\text{perp}$ changed from a continuous on-chain CLOB to a frequent batch auction (one uniform clearing price per market per block): the clearing price is computed identically by every validator in-state, so the κ basis has no single-proposer surface (the liability of an off-chain memclob, where the per-block mark is one proposer’s unattested read) and no continuous-CLOB MEV vector. §5.1 κ price-formation made explicit as benchmark-grade — stake-weighted-median index, TWAP-smoothed mark off the clearing price, clamping — mirroring post-LIBOR SOFR (transaction-anchored median + trim) and TradFi closing call-auctions; and the κ -yield-curve aggregation corrected to median implied yields at fixed tenors with on-chain re-fit (raw NSS parameters are non-identified and must not be medianed). All three changes are matching-agnostic w.r.t. the corpus — Papers 1/3/13 and the audited FundingEngine/OracleAdapter are unchanged.

v0.7.5 corrections (2026-06-12): Round-2 audit remediation — compliance language descoped to the riba layer corpus-wide (κ -curve/benchmark/instruments now “riba-free; product-layer Shariah questions remain with scholars”); funding spec unified to the 1-hour epoch ($\pm 0.75\%$ clamp) matching the audited FundingEngine; κ derivation unified to the mechanical $|\text{basis}| \times 8760$ rule ($\kappa_{\text{scaling}}/0.124$ removed); put price restated in P2’s two-region form; isomorphism table corrected to $\kappa - \iota \leftrightarrow \lambda$ ($\kappa \leftrightarrow \lambda$ at $\iota=0$); $\iota=0$ -in-consensus claim scoped to native primitives (EVM apps gated by $\times/\text{shariah}$ whitelist); R1–R4 product gates added to the $\times/\text{shariah}$ spec;

OMO transmission routed through the κ -yield-curve state (not the perp oracle); Phase 2 reordered asset-linked-first (κ -Gold, κ -Sukuk-yield lead; cash-settled perpetual remains subject to SBR-5); iCDS v1 premium-flow bug disclosed; micro-sukuk yield reframed as variable risk-sharing distributions; sim results restated with $R=30$ seeds; PMFBY framed as units-level agreement; $195 = 40+154+1$ municipal; SOFR-negative error fixed (euro-area/Japan); stale figures refreshed (13 contracts, 749/749 tests); references made two-author (2026a–e) + P10/P11 companions added; block-header residual purged; §4 renumbered.

v0.7.6 (2026-06-12): Added §5.2.2 — the Validator Liveness Sleeve (counter-cyclical tabarru pool for fee-only consensus under persistent macro winters): conjunction trigger (volume+ κ , both 24h-sustained, fixed governance constants), shortfall payout ($\beta=1.3$ dispersion allowance, performance-weighted, 2%/wk decay), 12-month cap with genesis seed, governance separation. Parameters validated by Monte-Carlo (sleeve_model/): 0% trigger false positives; censorship strictly dominated by the shortfall formula; severe-winter validator survival $84 \rightarrow 90\%$, black-swan $67 \rightarrow 78\%$.

v0.7.5b (2026-06-12): Final integrity sweep — remaining in-text solo-author companion cites unified to Ahmed & Bhuyan (2026a–e; 2026f remains single-author), and one residual §13 “Shariah-compliant by construction” phrase descoped to the riba layer.

v0.7.4 corrections (2026-06): External-audit remediation — unified κ publication to ABCI++ state (purged residual “block header” language); fixed the validator-economics arithmetic and the $0.083 \rightarrow 0.124$ κ -scaling collision; reconciled $\iota=0$ enforcement to the CheckTx-assert+slash model; excised all HyperEVM infrastructure reliance (Phase-2 now Arbitrum $\rightarrow$ native Cosmos x/evm); and moved NSS curve-fitting off-chain — validators fit locally and submit $\beta_0.. \tau_2$ via vote extensions, aggregated by stake-weighted median in PreBlocker (non-linear optimisation can’t run deterministically in a BFT EndBlocker). Added a §7 classical-foundations hook (al-Ghazālī / Ibn Taymiyyah / Ibn Khaldūn) pointing to the companion paper *Classical Foundations of the κ -Substrate*.

v0.7.3 additions (2026-06): Consolidated the §5.2 reactor-core architecture — rewrote x/perp (on-chain CLOB with inverted-key price-time priority, rolling-window TWAP mark, strict signed- F_t vs non-negative- κ separation, κ written to module state for the block.kappa precompile) and added the x/oracle module (ABCI++ vote-extension BFT stake-weighted-median index price, $\geq 2/3$ liveness threshold, staleness guard pausing κ derivation during outages).

v0.7.2 additions (2026-06): Added x/distribution Wakala fee override, macro-monetary implications, and finalized x/shariah module-level leverage caps.

v0.7.1 corrections (2026-06): Reconciled the empirical and pricing claims with the corrected SSRN papers — (i) §2.1 AHJ Proposition 3 price restated as $f = (\kappa - \iota) / (\kappa + r_b - r_a) x$; (ii) §2.5.3 and §6 κ -yield-curve evidence moved from the earlier calibrated-synthetic seven-market panel to the real 195-instrument USD-sukuk cross-section, and the circular SOFR Diebold–Mariano horse-race dropped; (iii) §3.4 iCDS riba relocated from the par spread (which is discount-independent and therefore riba-neutral) to the present value; (iv) §3.3 India takaful result marked a units-level coincidence, not a validation.

v0.7 additions from v0.6: (i) §1.4 new — “The project the mathematics made”: the narrative of how a perpetual futures DEX became a proof about the foundational axiom of financial mathematics, and how the competitive research revealed an ecosystem ready for exactly what the κ -Chain provides; (ii) §13 Conclusion rebuilt as five subsections — 13.1 What the mathematics built (κ -surface, not κ -rate), 13.2 What the L1 architecture changes (HAQQ as proof of the wrong-layer problem, $\iota=0$ substrate vs application-layer compliance), 13.3 What the competitive landscape revealed (IsDB, QFC, BNM ready; scaffolding already built), 13.4 The political economy conjunction (riba loading as systematic mispricing, correct pricing as the intervention), 13.5 What this is (the question the κ -Chain answers, and its destination).

v0.6 additions from v0.5: §5.3 Cosmos native EVM; §5.4 IBC v2 and QFC channel; §5.5 three-stage validator model (ICS $\rightarrow$ PoA $\rightarrow$ PoS); §10 Phase 3 roadmap; §11.2 IsDB PoU patent and QFC regulatory questions; §12 related work expanded.

v0.5 additions from v0.4: §4.1 hidden $r>0$ axiom in every L1; §4.4 Economic OS framing; §12 Avalanche, Circle Arc4, Hyperliquid precision; §13 two-axiom conclusion.

v0.4 additions from v0.3: life takaful example, purity summary table, practical synthesis paragraph in §2.5.2.

v0.3 additions from v0.2: historical accident framing, r vs κ comparison table, honest numerical caveat, Source 4 self-referential bootstrap, κ -surface naming.

v0.2 corrections from v0.1: SOFR contamination acknowledged, κ plurality clarified, fixed-income framing removed, logical dependency foregrounded.

Abstract

We propose the κ -Chain: a sovereign Layer 1 blockchain whose native monetary primitive is the convergence intensity κ , derived from the random stopping time representation of perpetual futures contracts (Akerer, Hugonnier & Jermann 2025), rather than the risk-free interest rate r that underlies every existing financial blockchain. The theoretical foundation rests on six working papers (Ahmed & Bhuyan 2026a–e; Ahmed 2026f) that establish: (i) the interest parameter ι is separable from and eliminable from no-arbitrage pricing (the Separation Theorem); (ii) κ is formally isomorphic to the hazard rate λ in the Duffie-Singleton credit framework, giving a complete riba-free pricing foundation for perpetual contracts, sukuk, takaful, and credit default swaps; (iii) the κ -yield curve can be estimated empirically from sovereign sukuk spread data across GCC and Southeast Asian markets; and (iv) these results are computationally validated against a live deployed protocol (Baraka Protocol, Arbitrum Sepolia).

The κ -Chain is not a blockchain that adds Islamic financial products as applications. It is a blockchain where $\iota = 0$ is the substrate — present in every transaction, every block reward, every pricing primitive — because the absence of interest is not imposed as a constraint but is the natural language of the pricing mathematics. We describe the full architecture, the validator economics, the monetary policy implications for central banks in OIC-member states, and a three-phase roadmap from the currently deployed Baraka Protocol v3 testnet to a sovereign Cosmos appchain. We document honestly what is built, what is proven, and what remains to be done.

Keywords: Islamic finance, κ -rate, convergence intensity, blockchain, perpetual futures, sukuk, takaful, iCDS, riba-free pricing, Layer 1, Cosmos SDK, AAOIFI, Shariah governance, monetary primitive

1. Introduction

1.1 The 50-year impasse

Islamic finance prohibits riba — the predetermined excess in exchange that arises from the time value of money. For fifty years, this prohibition has coexisted uneasily with modern financial mathematics, which takes the risk-free interest rate $r > 0$ as an axiom and derives all pricing from it. Bond prices, option prices, insurance premiums, credit default spreads — every instrument in the \$700 trillion derivatives market is a downstream consequence of that single axiom.

The unresolved tension has produced a paradox. The \$3.9 trillion Islamic finance industry exists, is growing at 10-15% annually, and serves 1.8 billion Muslims. But it operates without a rigorous mathematical foundation for credit pricing that does not require a positive interest rate. Practitioners use SOFR or conventional government bond yields as benchmarks even for ostensibly Islamic instruments — creating an implicit riba component in products whose explicit terms are structured to avoid it.

No amount of contractual engineering solves this problem. The riba is not in the contract terms. It is in the discount rate used to price the contract. A sukuk structured as a murabaha and priced using a SOFR-derived spread still contains riba at the pricing layer, regardless of its Shariah-compliant legal form. The Islamic finance industry has, for fifty years, been addressing the symptom while the cause remains in the mathematics.

1.2 The mathematical resolution

In 2025, Akerer, Hugonnier & Jermann published a proof in *Mathematical Finance* that a perpetual futures contract has a unique no-arbitrage price when the interest parameter ι is set to zero. The price is the expected value

of the underlying asset at the random stopping time when the contract converges — no discounting required:

$$f_t = \mathbb{E}_t^{Q_a}[X_{\theta_t}]$$

This result, combined with the formal isomorphism between the perpetual pricing framework and the Duffie-Singleton reduced-form credit model (established in Ahmed & Bhuyan 2026b), yields what we term the Separation Theorem: *interest is a separable, eliminable component of no-arbitrage pricing, not an intrinsic mathematical necessity*. The stopping time distribution carries all timing information previously attributed to the discount factor.

This is the mathematical statement Islamic finance has required but lacked for fifty years.

1.3 From theorem to infrastructure

A mathematical theorem, however precise, does not by itself change a financial system. The theorem becomes infrastructure when:

1. The rate κ derived from it can be estimated empirically from real market data (Paper 3, Ahmed & Bhuyan 2026c: a κ -yield curve built on a real instrument-level cross-section of 195 actively-quoted USD sukuk — 40 sovereign, 154 corporate, 1 municipal; Cbonds — plus roll-down-free constant-maturity κ histories for five issuers, 2018–2026).
2. The instruments priced by κ can be closed-form, practically computable, and riba-free — product-layer Shariah questions remain with scholars (Papers 2, 4, 5: Islamic perpetual sukuk, actuarially fair takaful, Islamic credit default swaps — all with on-chain implementations).
3. The mechanism that produces the primary κ signal — the convergence of perpetual futures prices to spot prices at $\iota = 0$ — can be deployed as a live protocol. This point is architectural, not commercial: the perpetuals DEX is not merely the first product. It is the mathematical prerequisite for the other three. The on-chain κ oracle that prices sukuk, takaful, and iCDS in their purest form can only exist if there is a live $\iota = 0$ perpetuals market generating the basis from which κ is observed. Every trade on Baraka makes the entire Islamic financial stack more accurate. (Baraka Protocol v3: 13 contracts, 749/749 tests passing across 14 audit passes, live on Arbitrum Sepolia since March 2026, IFA26 “Best Emerging Innovation” award.)
4. That protocol can become the consensus layer of a sovereign blockchain, so that κ is not a third-party oracle imported from outside but a quantity computed by the chain from its own activity and published with every block.

This white paper describes that blockchain — the κ -Chain.

1.4 The project the mathematics made

This project did not intend to be what it became.

Baraka Protocol started as a Shariah-compliant perpetual futures DEX — a product for Muslim traders who wanted to hedge without riba. Useful, limited, defensible. Then Paper 1 cited AHJ, and AHJ proved that $\iota = 0$ achieves unique no-arbitrage. Paper 2 proved the isomorphism between the perpetual pricing framework and the Duffie-Singleton credit model. The isomorphism yielded the Separation Theorem. And the Separation Theorem said something that no one in fifty years of Islamic finance scholarship had managed to say with mathematical precision: *interest is not logically necessary for financial pricing to work*.

At that point the project stopped being about a DEX. The DEX became evidence — the proof of concept that $\iota = 0$ operates correctly in a live market with real money. The claim the papers make is not about the DEX. It is about the foundational axiom of modern financial mathematics.

Every subsequent step — Papers 3, 4, 5, 6, the L1 architecture, the competitive landscape research — has been the implication of that theorem working itself out. Not ambition expanding the project. The mathematics expanding the project by following its own logic.

The competitive research made this visible in a specific way. HAQQ has \$400 million in funding, scholar endorsement, and a Shariah Oracle. It is among the best-funded Islamic blockchains to date. And it has a predetermined inflationary block reward that is structurally *riba*, sitting below the Shariah Oracle's visibility. \$400 million did not solve the problem because \$400 million was spent at the wrong layer — the application layer, not the monetary primitive layer. IsDB holds a US patent on a consensus mechanism that ties validation authority to network usage rather than capital. Qatar's QFC has regulatory infrastructure for Islamic finance blockchain compliance but no pricing engine to connect it to. Bank Negara Malaysia and SAMA run dual monetary systems and are looking for a market-determined Islamic rate to replace SOFR.

The ecosystem is ready. The institutions are ready. The regulatory pathways exist. What was missing was the mathematical foundation for a *riba*-free monetary primitive. Papers 1–6 provide exactly that. The κ -Chain is the infrastructure that implements it.

None of this was planned at the outset. All of it was implied by the first theorem.

2. Mathematical foundations

2.1 The AHJ framework

Let $(\Omega, \mathcal{F}, P)$ be a probability space with filtration satisfying the usual conditions. Let X_t denote the spot price of an underlying asset, and let Q_a be the long account's pricing measure. A perpetual futures contract on X_t with convergence parameter $\kappa > 0$ and interest parameter $\iota \geq 0$ is a continuously-settled contract with no fixed expiry.

Under constant parameters, Proposition 3 of Akerer et al. (2025) establishes that the unique no-arbitrage perpetual futures price is:

$$f_t = \frac{\kappa - \iota}{\kappa + r_b - r_a} \cdot X_t$$

When $r_a = r_b$ (symmetric funding rates), this reduces to:

$$f_t = \frac{\kappa - \iota}{\kappa} \cdot X_t$$

The convergence stopping time θ_t has conditional density under Q_a :

$$q(\theta_t \in ds | \mathcal{F}_t) = (\kappa_s - \iota_s) \exp \left(- \int_t^s (\kappa_u - \iota_u) du \right) ds, \quad s \geq t$$

The perpetual futures price satisfies (Theorem 3, Akerer et al.):

$$f_t = \mathbb{E}_t^{Q_a} \left[e^{-\int_t^{\theta_t} \iota_s ds} \cdot X_{\theta_t} \right]$$

The price is the discounted expected spot value at the random stopping time. The discount is ι — the interest parameter — not the funding rates r_a or r_b .

2.2 The critical case: $\iota = 0$

When $\iota = 0$, Theorem 3 of Akerer et al. establishes that no-arbitrage is satisfied. Substituting:

$$q(\theta_t \in ds | \mathcal{F}_t) = \kappa e^{-\kappa(s-t)} ds \quad \theta_t \sim t + \text{Exp}(\kappa)$$

$$f_t = \mathbb{E}_t^{Q_a}[X_{\theta_t}]$$

Three consequences are immediate:

No discounting. The expectation is taken without any interest discount. The price is the pure expected value of the spot at convergence.

κ carries all timing information. The distribution of when the contract pays off is governed entirely by κ . Larger κ means faster convergence; smaller κ means slower convergence. No interest rate is needed to encode timing.

$f(t) = X(t)$ at fair value. When funding rates are symmetric and the market is in balance, the futures price equals the spot price with no premium and no discount.

2.3 The Separation Theorem

The conventional wisdom in financial mathematics is that the interest rate is necessary for pricing time-dependent instruments — that discounting the future is mathematically required for fair valuation. The AHJ result, combined with the Duffie-Singleton isomorphism (§2.4), formally refutes this.

Theorem (Separation Theorem, Ahmed & Bhuyan 2026b). *Interest is a separable, eliminable component of no-arbitrage pricing. The stopping time distribution θ_t carries all timing information previously attributed to the discount factor $e^{-\int_0^t r \cdot dt}$. Setting $\iota = 0$ achieves unique no-arbitrage perpetual futures pricing; the formally isomorphic credit pricing framework achieves no-arbitrage at $r = 0$.*

The proof follows from the bijectivity of the isomorphism in §2.4 and the AHJ no-arbitrage result. The empirical corroboration is the European negative-rate environment (2014–2022), during which credit markets priced risk efficiently at $r \leq 0$ — discovering empirically what the isomorphism establishes theoretically.

The historical accident. Conventional finance fused two separate things into one parameter r : (1) the timing of when a payoff happens — when does the asset mature, when does default occur; and (2) the time value of money — the *riba* component, the predetermined reward for waiting regardless of economic outcome. These were always separable. The AHJ framework separates them. κ handles timing. At $\iota = 0$, the time value component is removed entirely. The price still works. No-arbitrage is still satisfied. The mathematics holds.

Interest is not a mathematical necessity. It is a historical accident — the way two distinct things got bundled into one parameter by the early architects of financial theory, and never unbundled until 2025. The Separation Theorem is the unbundling.

This theorem is the precise mathematical statement that Islamic finance has required for fifty years. *Riba* — the predetermined excess from the time value of money — is not a logical prerequisite for fair pricing of time-dependent financial instruments.

2.4 The Duffie-Singleton isomorphism

The reduced-form credit risk model of Duffie & Singleton (1999) prices a defaultable bond as:

$$P(t, T) = \mathbb{E}_t \left[e^{-\int_t^T (r_s + \lambda_s(1 - \delta_s)) ds} \right]$$

where λ is the default hazard rate, δ is the recovery rate, and r is the risk-free rate.

Proposition (Ahmed & Bhuyan 2026b). *The AHJ perpetual pricing formula and the Duffie-Singleton defaultable bond formula are formally isomorphic under the identification:*

AHJ quantity	Duffie-Singleton quantity
Convergence intensity $\kappa - \iota$	Default hazard rate λ

AHJ quantity	Duffie-Singleton quantity
Interest parameter ι	Risk-free rate r
Stopping time θ_t	Default time τ
Perpetual contract	Defaultable bond

(The AHJ stopping intensity is $\kappa - \iota$; at the riba-free point $\iota = 0$ this reduces to the headline identification $\kappa \leftrightarrow \lambda$ used throughout.)

Corollary. No-arbitrage at $\iota = 0$ in the AHJ framework implies no-arbitrage for credit pricing at $r = 0$. The stopping time distribution alone carries all timing information.

This isomorphism is the bridge that extends the AHJ result from perpetual contracts to the entire credit market — sukuk, takaful, iCDS, and any instrument whose pricing depends on a time-to-resolution distribution.

2.5 The κ -rate as Islamic monetary primitive

2.5.0 What κ is and what it is not Before the formula, a foundational distinction. Critics will observe that κ and r can produce similar numerical values in some markets. They will say: “It is the same thing with a different name.” This objection must be met directly, because the answer is the core of the Islamic finance case.

Property	Interest rate r	Convergence intensity κ
What it measures	Time passing — reward for waiting	Speed of a real event — market convergence, default, loss
Predetermined?	Yes — fixed at contract inception	No — implied continuously from live market conditions
Changes with the market?	No — charges regardless of outcome	Yes — updates as the real world updates
Compensates for	Lending capital over time	Real economic risk in a specific domain
Set by	Central bank committee	The domain’s own market or actuarial data
Islamic status	Riba — prohibited	Riba-free — no time premium (product-layer status with scholars)
Closest analogy	Rent on money	Insurance premium

The analogy that clarifies the distinction most sharply: a car speedometer and a heart rate monitor both measure rate — kilometres per hour and beats per minute — using the same mathematical concept of frequency. Nobody asks “what does my car speed have to do with my pulse?” They are measuring completely different phenomena using the same mathematical structure. BTC- κ and sukuk- κ are exactly this. The shared mathematics is structural, not substantive.

The honest caveat on numerical similarity: κ and r will sometimes produce similar numbers. A sovereign with $\kappa \approx 2.4\%$ and a conventional borrowing rate of $\approx 2.5\%$ might appear identical to a superficial reading. The response is that economic content differs even when numbers are similar — the same way a profit-sharing return and an interest payment can both be 8% per year while one is halal and one is haram. What determines Shariah status is not the magnitude but the structure: is the charge predetermined and independent of outcomes, or does it emerge from real economic conditions and vary with them? κ emerges and varies. r is predetermined and fixed.

2.5.1 The structural formula and its honest caveat Define the point estimator for κ from observed market data as:

$$\hat{\kappa} = \frac{s}{1 - \delta}$$

where s is the observed spread of an Islamic credit instrument (sukuk or iCDS) and δ is the recovery rate. This formula (derived in Ahmed & Bhuyan 2026b,c) connects the observable market spread directly to the convergence intensity without any reference to interest rates — the formula itself is *riba*-free.

However, a transparency requirement: the spread s used in the empirical work of Ahmed & Bhuyan (2026c) is sourced from existing sovereign sukuk markets, which currently price against SOFR or conventional government bond benchmarks. The Separation Theorem proves that s contains two bundled components — a pure credit risk signal (*halal*, what κ isolates) and a *riba* loading from the conventional discount rate (*haram*, what the formula discards). The input is currently contaminated; the extraction procedure removes the contamination.

This is a transitional condition, not a permanent feature. Section 2.5.2 describes the three sources of κ , in order of increasing purity, and how the architecture transitions across them.

2.5.2 Three sources of κ — from contaminated to pure Source 1: Existing sovereign sukuk spreads (available now, partially contaminated). $\hat{\kappa} = s/(1-\delta)$ applied to current market data extracts the pure credit risk signal from SOFR-tainted spreads. The Separation Theorem justifies the extraction mathematically. This is how Ahmed & Bhuyan (2026c) build the empirical κ -yield curves. It produces valid κ estimates today and provides the academic validation baseline, but acknowledges the input is not yet *riba*-free at source.

Source 2: Real-world event frequencies (available now, 100% pure). For non-financial instruments, κ can be calibrated entirely from actuarial and physical data with no connection to financial markets:

Instrument	κ source	Concrete example
Agricultural takaful	Historical crop loss frequency	Drought every ~12.5 years $\rightarrow \kappa \approx 0.08$ (measured $\hat{\kappa}_{MLE}$ 0.071–0.147 across South/Southeast Asia)
Life takaful	Age-adjusted mortality tables	Age 35 $\rightarrow \kappa_{mortality} \approx 0.002$ p.a.
Infrastructure sukuk	Engineering project timeline	7-year highway project $\rightarrow \kappa = 1/7 = 0.143$
Sovereign sukuk	Government project completion data	State-specific actuarial history

These κ values are derived from physical reality — meteorological records, actuarial tables, engineering reports — and contain zero *riba* by construction. No financial markets. No SOFR. No committee. The agricultural takaful study in Ahmed & Bhuyan (2026d) uses exactly this approach: κ calibrated from historical drought frequencies and compared against PMFBY (a units-level agreement, not a strict validation — see §3.3), with no conventional financial data anywhere in the computation. The life takaful figure ($\kappa_{mortality} \approx 0.002$ for a 35-year-old) comes directly from standard actuarial tables and has no relationship to any interest rate system.

Source 3: Live perpetuals market (available post-Baraka mainnet, 100% pure). Once the Baraka Protocol perpetuals market is live at $\iota = 0$, κ is observed directly from the on-chain basis:

$$\hat{\kappa}_t = \frac{|P_{\text{mark}}(t) - P_{\text{index}}(t)|}{P_{\text{index}}(t)} \times 8760 \quad (\text{1-hour epochs per year})$$

This computation involves no conventional financial market data, no SOFR, and no interest rate of any kind. It is generated by a market whose structural impossibility of $\iota > 0$ is enforced at the consensus layer. This is the purest observable κ possible — and it is the long-term source that the κ -Chain is built to produce.

Source 4: Self-referential Islamic market data (long-term, increasingly pure). As Islamic institutions begin pricing instruments using κ rather than SOFR — issuing sukuk on the κ -Chain, calibrating takaful premiums from actuarial data, settling iCDS at $s^* = \kappa(1-\delta)$ — the market data generated becomes progressively less contaminated.

The first sukuk priced natively on Baraka using κ as the benchmark creates a new clean data point. The second adds another. Over time a κ -yield curve is constructed entirely from riba-free transactions.

At that stage κ becomes self-referential: priced from a purely halal system that no longer needs SOFR as any input. The transitional dependency on contaminated data disappears not by policy decision but by the accumulation of clean market activity. Papers 2–5 build the methodology so that when the data exists, the construction procedure is already specified.

Summary of purity by source:

κ source	Purity	Available
Baraka perpetuals basis	100% pure	After mainnet launch
Real-world event frequencies (actuarial, engineering)	100% pure	Now
Sovereign sukuk spreads (current market)	Partially contaminated	Now — SOFR-tainted input
Future Islamic market data (self-referential)	Increasingly pure $\rightarrow$ 100%	As ecosystem grows

The practical synthesis: for the perpetual futures DEX, κ is 100% pure from day one — it comes from Baraka’s own market where $\iota = 0$ is immutable. For sukuk, takaful, and iCDS, the operational choice is between two already-pure sources: real-world event frequencies (available now, zero contamination) or Baraka’s own trading history (available post-mainnet, zero contamination). The contaminated sovereign spread data exists only in the academic papers as a validation baseline against 11 years of real market history (daily Cbonds G-spread histories, 2015–2026). It is not part of the protocol’s operational system.

2.5.3 Empirical properties of the κ -yield curve Using a real, instrument-level cross-section of 195 actively-quoted USD sukuk (40 sovereign, 154 corporate, 1 municipal; Cbonds), complemented by roll-down-free constant-maturity κ histories for five issuers (2018–2026), Ahmed & Bhuyan (2026c) characterises the κ -yield curve with the Nelson-Siegel-Svensson functional form and CIR mean-reversion dynamics:

$$d\kappa_t = \alpha(\bar{\kappa} - \kappa_t) dt + \nu\sqrt{\kappa_t} dW_t$$

Key empirical findings:

- κ is credit-quality ordered: median κ broadly increases with rating, from $\approx 0.9\%$ at AAA to $\approx 4.4\%$ at single-B, apart from the Saudi-dominated AA bucket, which prices just above single-A — read directly from the traded sukuk market.
- κ is non-negative by construction: the Feller condition $2\alpha\bar{\kappa} \geq \nu^2$ is satisfied in all twelve constant-maturity series (and in all twenty-one issuers of the full-universe panel). Unlike conventional benchmark rates — euro-area EONIA/€STR were negative 2014–2022 and Japan’s until March 2024 — κ cannot go negative — it is a probability intensity, and probabilities are non-negative by definition.
- κ mean-reverts with bias-corrected half-lives of 1.4–4.3 months at the well-sampled 7-year tenor — faster than the multi-quarter-to-multi-year half-lives documented for conventional sovereign spreads, consistent with credit information being incorporated efficiently. Identifiable constant-maturity dynamics extend to 21 issuers in the full 409-bond universe, with the Feller non-negativity condition holding in all of them.
- κ is upward-sloping by maturity: the investment-grade term structure slopes up at +0.63 bp/yr with rating fixed effects (+1.17 bp/yr pooled), confirmed at full-universe scale (+3.4 bp/yr under issuer and month fixed effects across 95 issuers and 165,557 bond-days) — a term premium with no interest-rate component.

The κ -yield curve is therefore a theoretically rigorous, empirically grounded (on real traded sukuk), non-negative, riba-free substitute for conventional risk-free curves across all Islamic credit pricing — product-layer Shariah questions remain with scholars — and one whose purity increases over time as the Baraka perpetuals market matures into the primary source.

3. The instrument families

The κ -rate, once established as a primitive, prices four complete instrument families without any interest component. All four are deployed as smart contracts in the Baraka Protocol v2/v3 codebase.

3.0 A critical clarification: there is no single κ

Before describing the instruments, a misconception must be prevented. κ is not a single rate the way SOFR is a single rate set by a committee and applied uniformly across instruments. There is no “the κ -rate” the way there is “the Fed Funds rate.”

κ is a methodology — a mathematical framework for measuring the intensity of any random resolution event — and every instrument calibrates its own κ from its own data source:

Instrument	κ measures	Calibrated from
BTC perpetual	Speed of BTC futures converging to BTC spot	Live on-chain basis: $(P_{\text{mark}} - P_{\text{index}}) / P_{\text{index}}$
ETH perpetual	Speed of ETH futures converging to ETH spot	Live on-chain basis, independently
Agricultural takaful	Crop loss event frequency	Meteorological/actuarial records
Infrastructure sukuk	Project completion intensity	Engineering timelines and government data
Sovereign sukuk	Sovereign credit event intensity	Market spreads or historical default data
iCDS	Reference entity default intensity	Historical default rates

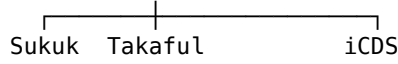
BTC- κ has no direct relationship to sukuk- κ . They share a mathematical structure — both are exponential stopping time intensities describing “how fast does something happen” — but they are calibrated independently from entirely different data sources. A Baraka trader’s BTC position does not affect the κ used to price a Malaysian sovereign sukuk.

This plurality is actually stronger than the conventional system, where everything anchors to a single SOFR number controlled by the Federal Reserve. In the κ framework, each instrument is priced from its own real economic risk, not from US monetary policy. Conventional finance has a yield curve — one interest rate per maturity, for one benchmark. The κ framework produces a κ -surface: different convergence intensities for different assets, different instruments, and different maturities. Each cell of that surface is calibrated independently from its own domain’s data. No single committee controls any of it.

What Baraka Protocol specifically provides for the non-perpetuals instruments: Baraka does not use BTC- κ to price sukuk. What Baraka provides is three things: (1) a proof of concept that the κ framework operates correctly in a live market with real money at stake; (2) a legitimacy signal — if κ prices BTC perpetuals correctly, the identical mathematics prices sukuk and takaful correctly; and (3) the long-term primary source of the κ -surface, as Baraka’s trading history accumulates into the cleanest available Islamic market data. The sukuk products use their own κ calibrated from project timelines or sovereign spread data — independent of BTC — but validated by the same mathematical framework that Baraka proves works in practice.

The perpetuals DEX is not one product among four. It is the mathematical prerequisite for the others. The logical dependency is architectural:

Perpetual Futures DEX (Baraka)
↓
Observable on-chain κ (live, pure, riba-free)
↓



Without a live perpetuals market at $\iota = 0$, the on-chain κ oracle has no primary source. The other instruments can bootstrap from actuarial data and sovereign spreads (§2.5.2 Source 1 and Source 2) — but the long-term architecture requires the perpetuals market as the foundation. This is not a business decision about launch sequencing. It is a mathematical requirement.

3.1 Perpetual futures ($\iota = 0$ mechanism)

The FundingEngine implements:

$$\text{rate per 1-hour epoch} = \frac{P_{\text{mark}} - P_{\text{index}}}{P_{\text{index}}}, \quad \text{clamped to } \pm 0.75\%$$

The funding payment when the perpetual price deviates from spot is:

$$\text{payment} = \text{position size} \times \text{rate} \times \text{elapsed epochs}$$

This payment is not predetermined. It is not independent of economic conditions. It is zero when mark equals index. It reverses direction when market conditions reverse. None of the three El-Gamal (2006) conditions for *riba* are met: the payment is not predetermined, not independent of outcomes, and not labeled as interest. This is the empirical basis for the Shariah analysis in Paper 1.

3.2 Islamic perpetual sukuk

A preliminary clarification on investor returns: perpetual sukuk are not fixed-income instruments. Fixed income — a predetermined return on capital over time — is structurally *riba al-nasi'ah* regardless of how the cash flows are labelled. The Islamic perpetual sukuk described here is a risk-sharing investment certificate whose returns are variable, asset-linked, and *mudarabah*-compatible.

What investors actually receive: The investor contributes capital backed by a real asset with value X_t . They receive periodic profit distributions structured as *ijarah* — a share of the economic output of the underlying asset, not a predetermined interest rate. At the random redemption time θ_t , they receive the fair market value of the asset. If the asset has appreciated, they participate in the upside. If it has declined, they bear that risk. This is risk-sharing, not lending at interest.

From the isomorphism (§2.4), the no-arbitrage price of such a certificate — an instrument structured per AAOIFI Standard No. 17 (certification rests with a Shariah board) with no fixed maturity — is:

$$P_{\text{sukuk}}(t) = \mathbb{E}_t^{Q_a}[X_{\theta_t}]$$

where θ_t is the redemption stopping time with intensity κ_{sukuk} , calibrated from the specific asset's project timeline or sovereign spread data (not from BTC- κ or any perpetuals market). This is the expected fair-market value of the underlying asset at the random redemption event — no interest discount, no predetermined coupon, no *riba*.

The closed-form expression under CIR stochastic- κ dynamics yields the affine survival probability:

$$S(0, T) = A(T) e^{-B(T)\kappa_0}$$

where $A(T)$ and $B(T)$ are the standard CIR bond-price coefficient functions (not to be confused with the *takaful* benefit B of §3.3). This is the Islamic equivalent of the zero-coupon bond price in the CIR term structure model — a term structure of risk-sharing returns, not a term structure of interest.

The correct pitch to investors: “Equity-like returns from real asset exposure — you share in asset upside via ijarah profit distributions and market-value redemption. The return is variable, tied to asset performance, and priced using κ derived from the asset’s own risk characteristics. No predetermined rate. No riba.”

3.3 Takaful (mutual Islamic insurance)

The riba-free takaful premium for a contract with benefit B and hazard intensity κ is (Ahmed & Bhuyan 2026d):

$$\pi^* = \kappa \cdot B$$

Under CIR stochastic dynamics:

$$\pi_{\text{stoch}}^*(T) = \mathbb{E}_0[\kappa_T] \cdot B = [\bar{\kappa} + (\kappa_0 - \bar{\kappa})e^{-\alpha T}] \cdot B$$

Empirical illustration: For India’s PMFBY national crop insurance program, the model-implied $\hat{\kappa} = 12.06\%$ is numerically close to the actuarial rate ($\approx 12\%$) — but this is a units-level coincidence between a loss frequency and a premium rate, not a strict validation: the identical $\hat{\kappa}$ for Pakistan misses its observed rate by $1.6\text{--}3.4\times$ (Ahmed & Bhuyan 2026d). Conventional crop insurance embeds a riba loading of:

$$\Pi_{\text{riba}} = \frac{\kappa B r}{1 + r}$$

At $r = 8\%$, this equals 7.4% of the riba-free premium — extracted annually from subsistence farmers who have no awareness that the discount rate in their insurance contract constitutes a systematic transfer.

The EverlastingOption contract in Baraka Protocol implements the closed-form put price from Proposition 6 of Akerer et al.:

$$\Pi_{\text{put}}(x, K) = \begin{cases} C_p x^{\beta_p}, & x \geq K \\ (K - x) + \frac{K^{1-\beta_c}}{\beta_c - \beta_p} x^{\beta_c}, & x < K \end{cases} \quad C_p = \frac{K^{1-\beta_p}}{\beta_c - \beta_p}$$

where $\beta_p = \frac{1}{2} - \sqrt{\frac{1}{4} + \frac{2\kappa}{\sigma_X^2}} < 0$ and $\beta_c = \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2\kappa}{\sigma_X^2}}$ are the two roots of the characteristic equation (value-matching and smooth-pasting at $x = K$) — the two-region form of Ahmed & Bhuyan (2026b); a single power law extended through $x < K$ would omit the intrinsic value and misprice the floor.

3.4 Islamic Credit Default Swap (iCDS)

The fair iCDS spread at $t = 0$ is (Ahmed & Bhuyan 2026e):

$$s^* = \kappa(1 - \delta)$$

A standard property of flat-intensity credit pricing is that this par running spread is independent of the discount rate ι : both the protection and premium legs terminate at the default time, so the discount factor cancels between them. A correctly-priced *conventional* CDS therefore quotes the same fair spread $s_{\text{conv}} = \kappa(1 - \delta) = s^*$ at every maturity — the par spread is automatically riba-neutral.

The riba is real, but it lives in present-value terms, where the discount does not cancel. The present-value riba premium — the discounting haircut applied to the future loss — is

$$\Pi_{\text{riba}}^{\text{PV}} = (1 - \delta) \frac{\iota}{\kappa + \iota},$$

which vanishes at $\iota = 0$, increases monotonically in ι , and stays under 4% of notional at realistic 5–10 year sukuk maturities (Ahmed & Bhuyan 2026e). The iCDS removes it both by pricing at $\iota = 0$ and structurally — collateral held in a non-interest-bearing tabarru pool, full collateralisation, and an objective on-chain credit event. The empirical analysis runs on 62 real sovereign USD sukuk (Cbonds): $\hat{\kappa}$ read from sukuk spreads agrees with same-issuer conventional bonds within 6–16 bp and co-moves with sovereign CDS at correlation 0.97.

The corrected iCDS.sol contract on Arbitrum Sepolia (0x77a9878512744F59CA02B009bA92e29A59eb521b, deployed and verified 13 June 2026) implements the full open $\rightarrow$ accept $\rightarrow$ settle lifecycle at a Foundry-measured $\approx 897,000$ gas for one-year protection with three premium payments — a few US cents at L2 gas prices. (A v1 implementation bug — the premium quoted as the put *price* per interval rather than the fair $\kappa \cdot \Pi \cdot \Delta t$ premium *flow*, a $\sim 25\times$ overcharge — was disclosed in the June 2026 revision of Ahmed & Bhuyan (2026e) and fixed in this deployment; the κ -Chain x/credit design specifies the same corrected flow.)

4. Why a new blockchain is required

4.1 The hidden axiom in every existing L1

A Layer 1 blockchain is the foundational network that processes and finalises transactions independently, without relying on another network for security or consensus. It establishes the core rules, security mechanisms, and native token for an entire ecosystem. Whatever is encoded in those core rules is inherited by every application, every smart contract, and every financial instrument built above it.

This creates a problem that no amount of application-layer engineering can solve: every existing Layer 1 encodes the axiom $r > 0$ in its substrate — without intending to, without stating it, and without most of its builders realising it.

The evidence is in three structural features shared by every major chain:

Inflationary block rewards. Ethereum’s proof-of-stake issues new ETH to validators. Cosmos Hub’s staking yields are inflation-funded. BNB Chain, Solana, Avalanche — all pay validators a predetermined return for the passage of time, regardless of whether real economic activity occurred. This is structurally *riba*: a fixed reward for waiting, charged against non-stakers through dilution.

Fee assets priced at $r > 0$. Gas fees are paid in ETH, SOL, ATOM — assets whose markets price them using conventional discount rates. Every transaction on these chains implicitly references a pricing system built on $r > 0$, even when the transaction itself is structured to be Shariah-compliant.

No native concept of κ . The EVM has no opcode for convergence intensity. The Cosmos SDK has no module for stopping time intensity. These concepts do not exist in the state machine. An Islamic financial application built on any existing chain must import κ from outside — making it an oracle, not a substrate property.

The consequence: Shariah compliance on any existing chain is a feature applied over an $r > 0$ substrate. Features can be toggled, upgraded, circumvented, or governed away. The substrate cannot. The *riba* is structural, not incidental.

4.2 The oracle problem

On every existing blockchain — Ethereum, Arbitrum, Cosmos, Solana — a rate like κ would be delivered via an oracle: a third-party data feed that applications consume. This creates two structural problems for the Islamic L1.

The trust problem. An oracle can be manipulated. A validator set that controls a staked oracle can be bribed, coerced, or simply wrong. If κ is imported from outside, no market participant has a chain-level guarantee that the κ they are pricing against reflects actual market conditions.

The substrate problem. More fundamentally: if κ is an oracle, then the chain is not a κ -native system. It is a conventional chain with a κ feed bolted on. Every block reward, every gas fee, every settlement mechanism on

the chain is still structured around implicit interest assumptions inherited from the EVM or Cosmos SDK defaults. The Shariah compliance is an application-layer feature, not a substrate property. This is the current state of all “Islamic DeFi” — including Baraka Protocol on Arbitrum. It is Phase 1, not the destination.

4.3 The κ -Chain difference

On the κ -Chain, κ is computed from the perpetuals basis on-chain and published by the consensus layer as a consensus-validated state primitive via ABCI++ (see §5.1), exposed to the EVM through the `block.kappa` precompile. The computation is:

$$\hat{\kappa}_t = \frac{|P_{\text{mark}}(t) - P_{\text{index}}(t)|}{P_{\text{index}}(t)} \times 8760$$

the mechanical annualisation of the 1-hour funding epochs (§5.2). The cadCAD simulation of Ahmed & Bhuyan (2026f) demonstrates that $\hat{\kappa}$ is computable from protocol observables under adversarial-trader conditions; its simulated level is environment-specific and is not a market calibration (distinct from the agricultural-takaful intensities of §2.5.2).

The consensus-validated κ state (read via the `block.kappa` precompile) includes: - `kappa_current`: instantaneous κ computed from current perp basis - `kappa_ema_8h`: 8-hour exponentially weighted moving average (over the 1-hour funding samples) - `kappa_yield_curve[6]`: NSS parameters $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ — fit off-chain by each validator and aggregated by stake-weighted median in PreBlocker (§5.1), never optimised on-chain

Every smart contract on the chain can read `block.kappa` the same way Ethereum contracts read `block.timestamp`. No oracle. No trust assumption. κ is as native as block time.

4.4 What changes when $\iota = 0$ is in consensus

When $\iota = 0$ is enforced at the consensus layer — not by a smart contract guard but by the validator set’s transaction inclusion rules — no *native* pricing primitive on the chain carries an interest parameter — ι in the consensus state is pinned to zero. An application deployed in the EVM environment could still implement interest-like logic internally; such applications are excluded from the Shariah-certified instrument whitelist by the `x/shariah` gate rather than by the ι -assert, which governs the native modules. A lending protocol that tried to charge a predetermined interest rate would produce transactions whose decoded state transition implies a non-zero ι — and the `CheckTx` ante-handler asserts $\iota = 0$, so those transactions are rejected at the inclusion stage, and a proposer that nonetheless includes one is slashed. Interest is not a usable degree of freedom on the chain: ι is pinned to zero by consensus, not merely discouraged by a rule.

4.5 The κ -Chain as economic operating system

Circle, in launching Arc4 in 2026, described their Layer 1 as the “Economic Operating System for the internet” — a chain built not as a neutral substrate but as infrastructure that encodes specific financial rules at the base layer, making those rules available to every application without additional configuration.

The framing is exactly right, applied to the wrong axiom. Circle’s OS encodes the conventional financial system’s rules at the L1 level. The κ -Chain encodes a different axiom: $\iota = 0$.

An operating system determines what applications can do. Windows applications can address the Windows API. iOS applications can address the iOS API. Neither can address the other’s. An L1 is the financial OS: applications built on it can price instruments using the primitives the chain makes available. On Ethereum, those primitives include $r > 0$ staking yield, ERC-20 tokens priced in conventional markets, and gas fees denominated in an inflationary asset. On the κ -Chain, those primitives include `block.kappa`, the `x/takaful` pool, the `x/credit` sukuk module, and a block reward structure that pays from real activity rather than inflationary emission.

The 2026 L1 landscape has fragmented into specialised chains — performance (Solana, Hyperliquid), settlement (Ethereum), compliance (Avalanche subnets), payments (Circle Arc4). In a specialised multi-chain world, the

question for any new L1 is not “can we beat Ethereum?” but “which role does no existing chain fill?” The answer for the κ -Chain is unambiguous: no existing chain encodes $\iota = 0$ at the consensus layer, has a native κ oracle in its consensus state, or provides a complete Islamic financial instrument suite as native primitives. The role is vacant. The mathematics to fill it exists. The deployed protocol to prove it works already runs.

The difference between Shariah compliance as a feature and Shariah compliance as a substrate is the difference between a door with a lock and a wall. The Spring-Loaded Door framework identifies τ_{eff} — the decay timescale of reform — as inversely related to how deeply a change is embedded in the substrate. A constitutional amendment decays more slowly than a regulation. A substrate-level absence decays not at all.

5. Architecture specification

5.1 Consensus layer

Consensus mechanism: CometBFT (formerly Tendermint BFT). Byzantine fault-tolerant, deterministic finality, 1-3 second block time. The same consensus used by Cosmos Hub, dYdX v4, and 100+ production chains.

Consensus-validated κ state (ABCI++ vote extensions): The κ primitives are published as a consensus-validated *state* primitive via ABCI++, not as new block-header fields. Modifying the CometBFT block header is avoided deliberately: the header is a fixed protobuf that cannot be extended without forking consensus (perpetual security-patch debt) and would break IBC light-client compatibility. Instead, validators commit the oracle index price during consensus through vote extensions; the x/perp EndBlocker derives κ deterministically from the vote-extension-aggregated price and writes it to module state each block. The EVM accesses it natively through the `block.kappa` precompile (IKappaOracle). The published state schema:

```
KappaState (x/perp KVStore – written every block, read via block.kappa precompile) {
  kappa_current:      uint64 // annualised convergence intensity, 18-decimal, non-negative
  kappa_ema_8h:       uint64 // 8-hour EWMA over 1-hour samples (low-pass for macro instruments)
  kappa_yields:       bytes  // per-tenor median implied yields (the identified quantity)
  kappa_nss_params:   bytes  // NSS curve  $\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2$  – re-fit on-chain from kappa_yields
  iota:              uint64 // always 0; included for auditability
  shariah_epoch:      uint32 // increments when Shariah board updates parameters
}
```

Funding settles on a 1-hour epoch (matching the audited FundingEngine.sol reference contract, and aligned with dYdX v4 / Hyperliquid velocity rather than the legacy 8-hour Binance convention). The `kappa_ema_8h` field is a deliberate low-pass filter — an 8-hour EWMA computed over the 1-hour samples — so that slow-moving x/credit and x/takaful instruments price off a smoothed κ rather than high-frequency basis noise.

κ price-formation is benchmark-grade. The κ basis is $(\text{mark} - \text{index})/\text{index}$, where the mark is a TWAP over the per-block uniform clearing prices of the x/perp frequent batch auction (§5.2) and the index is a stake-weighted median of validators’ price submissions ($\geq 2/3$ power), both legs clamped and the index cross-checked across sources. This is the discipline that produces robust reference rates outside crypto: post-LIBOR SOFR is a volume-weighted *median of transactions* with outlier *trimming* (LIBOR failed precisely because it was a quote/submission average a single party could move); the IOSCO benchmark principles rank concluded transactions above quotes; and TradFi exchanges form their *official* reference price (the close) with a uniform-price call auction — exactly the on-chain FBA used here. κ is therefore never a single raw block print: it is a transaction-anchored, median-attested, TWAP-smoothed, clamped quantity — which is what makes it safe to expose as the chain’s monetary primitive even in thin markets (the §11.2 backstop — a *continuous gated blend* into the sukuk κ -yield curve as on-chain depth thins, not a manipulable hard switch — completes this).

The yield curve is aggregated in yield space, not parameter space. Running a non-linear NSS optimisation inside the BFT EndBlocker would be non-deterministic across node architectures and would halt consensus, so the fit is done off-chain — but the raw NSS parameters must not be medianed. The Nelson–Siegel–Svensson vector is *non-identified*: when $\tau_1 \approx \tau_2$ only $\beta_2 + \beta_3$ is pinned down (a flat likelihood along the $\beta_2 \leftrightarrow \beta_3$ direction), so a per-parameter median of independently-fitted curves can yield a curve that fits none of the inputs.

Instead, each validator evaluates its locally-fitted curve at a fixed tenor grid (e.g. 1m, 3m, 1y, 2y, 5y, 10y) — fitting with a *fixed* decay λ (Diebold–Li), making the local fit a unique deterministic OLS — and submits those implied yields in its ExtendVote payload alongside the index price. PreBlocker takes the per-tenor stake-weighted median (the same deterministic median used for P_index) and re-fits NSS on-chain to the agreed yields, writing kappa_yield_curve[6] to state. Aggregation thus operates on the identified, observable quantity (yields at tenors), never on the unidentified parameters, and the result is identical on every honest node.

$\iota = 0$ enforcement: The mempool validation module rejects any transaction whose decoded state transition would produce a non-zero ι value in the FundingEngine state. The check is $O(1)$: the ι parameter in the FundingEngine state (consensus-validated via ABCI++) is asserted to be zero on every block production. A proposer that includes $\iota > 0$ is slashed.

Block reward model: No inflationary emission. Block proposers earn 100% from protocol fees collected during the block. Fee distribution: - 60% $\rightarrow$ proposer validator - 20% $\rightarrow$ InsuranceFund (tabarru-structure: pooled, not owned by any party) - 20% $\rightarrow$ Shariah Board operational treasury

This is not a conventional proof-of-stake reward structure. It is a wakala model: the validator is an agent earning a disclosed fee for a defined service, with the remainder going to the mutual pool. The parallels to TakafulPool.sol wakala fee design are direct.

5.2 Native primitive modules (x/ modules)

The Cosmos SDK’s module architecture allows custom state machines to be composed into a chain. Each x/ module defines its own message types, state transitions, and query interfaces. The κ -Chain includes six native modules that are not applications running on top of the chain — they are part of the chain itself.

x/perp — Perpetual futures engine - On-chain frequent batch auction (FBA), not a continuous CLOB and not an AMM (AMMs cannot price high-leverage perpetuals without disastrous slippage). Orders accrue in the resting book during the block (sealed, with no intra-block time priority); the EndBlocker clears each market at one uniform price — the price that maximises crossing quantity, midpoint on ties. This is deliberate: the uniform clearing price is a pure function of the order set, computed identically by every validator in-state, so (a) the κ basis derives from one unambiguous, replayable mark with no single-proposer surface — the liability of an off-chain memclob (e.g. dYdX v4), where the per-block mark is one proposer’s unattested read of its own RAM book — and no continuous-CLOB MEV/front-running vector, since a sealed batch clearing at a single price eliminates it structurally (cf. Budish–Cramton–Shim); and (b) one clearing pass writes net fills rather than streaming continuous matches, minimising IAVL write churn. A ~1-second BFT block already batches, so continuous matching buys no real latency for the institutional users this market serves, and price-time priority adds nothing. The resting book uses dense integer-tick byte-keys (prefix || asset || priceTicks || orderID; raw 10¹⁸ WAD overflows uint64), with per-market order caps and aggressive pruning. (Production precedent: Injective’s on-chain FBA; the module is kept deliberately minimal — uniform clearing + basis emission — to avoid the maintenance burden that led Sei to deprecate its sprawling native CLOB.) - $\iota = 0$ enforced by the module state machine (not governance-adjustable) - Mark price = a rolling-window TWAP over the per-block FBA uniform clearing prices (a bounded window fully evicts a manipulated print once it ages out, unlike an EMA’s decaying tail; this reproduces the audited Solidity FundingEngine/OracleAdapter, whose 30-minute TWAP + thin-data fallback this module mirrors). Each block’s clearing price is folded into a persistent ring buffer and the TWAP is cached; when fewer than the minimum observations exist, the mark falls back to the index and κ is flagged degraded. - Signed funding F_t and non-negative intensity κ are strictly separated. Each block the EndBlocker computes the basis (mark – index)/index, clamps it to $\pm F_{\max}$ (a volatility circuit breaker, never an interest floor) to obtain the signed F_t used for mark-to-market settlement, then derives $\kappa = |F_t| \times \text{intervals_per_year}$ (8760 at the 1h epoch) as a strictly non-negative stopping-time intensity. The two are never conflated. - κ written to module state, not the block header. The EndBlocker writes kappa_current/kappa_ema_8h to the x/perp KV store; the EVM reads them through the block.kappa precompile (IKappaOracle), as 18-decimal WAD values. (See §5.1 on why this is ABCI++ state, not a header field.) - Funding settlement: every 1-hour epoch (matching the audited FundingEngine reference; the 8-hour EWMA kappa_ema_8h smooths these 1-hour samples for slow macro instruments), settled at the stored start-of-period rate over the elapsed (capped) intervals to defeat retroactive manipulation. Net transfer = $\Sigma(\text{position} \times \text{premium_rate})$. - All position and order-book state stored in the

module's KV store; no separate database

x/oracle — Byzantine-fault-tolerant index price (ABCI++ vote extensions) - The consensus engine is the oracle: each validator injects its index-price reading into its precommit vote extension, so price updates ride inside consensus and bypass the mempool entirely — there is no front-running or gas-griefing surface for oracle updates. - Five deterministic phases: ExtendVote (node-local fetch — the only non-deterministic step) → VerifyVoteExtension (lenient structural check; an empty extension is a valid abstention) → PrepareProposal (the proposer injects the prior height's ExtendedCommitInfo as the first transaction) → ProcessProposal (peers validate $\geq 2/3$ stake signed) → PreBlocker (deterministic aggregation). - P_index is a stake-weighted median with a deterministic total order (price, then validator address — an unstable sort here would split consensus), requiring $\geq 2/3$ of total network voting power to have submitted before the price is trusted. To move the median an attacker must control $>50\%$ of *participating* stake ($\approx >1/3$ of total if only the quorum votes), and BRKX carries no staking yield, so that stake is pure capital cost. This bound is simulation-validated: a Monte-Carlo Byzantine oracle in which the adversary captures the largest stakers and submits a +500 bp lie holds the stake-weighted median to 2–4 bp of bias for all $f \leq 1/3$ (a stake-weighted *mean* would be dragged hundreds of basis points), and a *withholding* attack past 1/3 trips the staleness guard into a safe pause rather than an accepted bad κ — a liveness fault, not a safety break (Paper 3 §“Systemic and adversarial robustness,” `simulations/byzantine_oracle/`). - Staleness guard: if no BFT quorum has updated an asset within MaxIndexStalenessBlocks, the price is marked stale and the x/perp EndBlocker pauses κ derivation for that market — a stale index against a fresh mark would otherwise fabricate basis. (The mark TWAP also pauses during the outage: trading without a live index is presumed distressed or manipulated.)

x/shariah — Compliance layer - Asset whitelist maintained by Shariah board multisig - Maximum leverage per market (genesis default 5x; board-adjustable downward to 1x, and never above a 10x hard ceiling enforced at the module level) - IPFS CID references for fatwa documents, with 3-of-5 board quorum for updates - Emergency halt: board can freeze x/perp and x/credit module state transitions immediately, without DAO vote - validate-Order() called by x/perp before every fill inclusion - R1 — asset-backing whitelist (existential gharar): tradable instruments must reference a whitelisted token over a real, registered, deliverable asset, not a bare price feed (per the companion Shariah-boundaries note, Ahmed & Bhuyan 2026g) - R2 — possession-gating (qabd hukmi): forward/sell orders rejected unless the party holds the asset-token (cryptographic control + registry record) - R3 — insurable-interest / no-naked rule (maysir): iCDS protection buyers must hold the reference obligation; sellers post full notional - R4 — expiry/delivery and tabarru wrappers: expiring deliverable structures and tabarru pools preferred over perpetual cash settlement for the certified class

x/credit — Sukuk and iCDS primitives - Perpetual sukuk issuance: sovereign or corporate issuers register, price in κ - iCDS market: open → accept → settle lifecycle, $s^* = \kappa(1-\delta)$ pricing - Settlement in USDC or PAXG (Shariah-board approved collateral)

x/takaful — Mutual insurance pools - Pool creation with parametric trigger specification - Tabarru contribution pricing: $\pi^* = \hat{\kappa} \cdot B$, where $\hat{\kappa}$ is calibrated from actuarial/physical data (drought frequency, mortality tables — §2.5.2 Source 2, the purest κ , with no perpetuums-market input); the x/perp κ benchmark is used only for market-linked covers - Claim settlement: automated when parametric index triggers - Agricultural pools: World Bank yield index or satellite data as trigger oracle - Wakala fee: 10% to pool operator, remainder to tabarru pool

x/gov — Dual-track governance - Track 1 (DAO): BRKX token holders propose, vote (4% quorum, simple majority), 48-hour timelock - Track 2 (Shariah board): 72-hour veto window; vetoed proposals cannot be executed; veto is irrevocable - Parameters that Shariah board controls exclusively: ι (always 0), asset whitelist, leverage caps, wakala fee bounds - Parameters that DAO controls (subject to board veto): fee distribution ratios, chain upgrade proposals, validator set expansion criteria

5.2.1 Fee distribution (x/distribution override)

The wakala block-reward model of §5.1 cannot be implemented with the stock Cosmos SDK x/distribution module, because that module is structurally interest-bearing in two ways. First, it assumes an inflationary source: the x/mint module credits newly minted tokens to the fee collector every block before distribution, and that minted portion is a predetermined, time-based return — *riba al-nasi'ah* embedded in the substrate itself. Second,

its allocation logic adds a performance bonus and a community tax on top of a power-weighted split. The κ -Chain replaces both.

No inflationary source. The `x/mint` module is removed from the application (or installed as a no-op minter that returns zero). The only balance present in the fee-collector account at the start of any block is the set of real transaction fees that users actually paid in the previous block. An empty block distributes nothing; there is no emission to fall back on.

Wakala allocation. A replacement `AllocateTokens` routine drains the fee-collector account each block into three receive-only module accounts in the fixed ratio of §5.1:

<code>proposer</code>	: 60%	→ block-producing validator (disclosed agency fee)
<code>InsuranceFund</code>	: 20%	→ <code>tabarru</code> pool (mutual; owned by no party)
<code>ShariahTreasury</code>	: 20%	→ Shariah board operational treasury

There is no proposer bonus and no community tax — the validator receives a single, disclosed fee for a defined service (wakala), with the remainder accruing to the mutual pool and the board treasury. The three ratios are governance parameters the DAO may tune, but they are bounded by an invariant the Shariah board controls: the ratios must sum to exactly one, and the combined `InsuranceFund` + `ShariahTreasury` share is floored, so the DAO cannot route the mutual pool's entitlement to validators by ordinary vote.

Determinism and integrity. The split uses integer (minor-unit) arithmetic only; the truncation remainder (“dust”) is assigned deterministically to the `InsuranceFund`, so every minor unit is accounted for, none is created, and the fee-collector balance is exactly zero after distribution. The routine reads only in-state values (parameters and account balances) — no floating point, no wall-clock, no oracle reads — so every honest validator computes an identical post-state, preserving BFT determinism. The `InsuranceFund` account is granted no staking or minting permission: it can hold contributions and pay parametric claims but can never earn a return on its balance, closing the back door by which a pooled fund could otherwise re-introduce interest. This mirrors the wakala fee design already proven in `TakafulPool.sol` at the EVM layer.

Macro-monetary implications. By encoding this Wakala split directly into the `x/distribution` state machine — bounded by the board invariant — and eliminating `x/mint`, the architecture achieves three systemic advantages. First, it eliminates the regressive wealth pump of standard Proof-of-Stake, where inflationary rewards dilute holders to enrich staking cartels; validators are instead forced to optimize for network utility and transaction throughput. Second, the automated routing to the `InsuranceFund` ensures self-capitalizing safety: the network taxes its own velocity to underwrite its own solvency, organically deepening the systemic backstop as derivatives volume scales. Third, routing block fees to the Shariah Board treasury provides native compliance funding, guaranteeing the scholars' financial independence from off-chain corporate payrolls and perfectly aligning with their irrevocable on-chain veto authority.

5.2.2 Validator liveness under stress: the `tabarru` sleeve

A fee-only consensus layer has no emission to fall back on: validator income is real transaction fees, and the dangerous regime is not the crash (liquidations and de-risking *raise* fee income in the acute phase) but the winter — the long low-volume aftermath of a persistent macro shock. The companion DSGE's central buffer result reappears here at the consensus layer: it is the *length* of the stress, not its depth, that drains any finite reserve. An undercapitalized validator set in month six of a winter is a censorship and liveness risk to the chain itself.

The architecture answers with a Validator Liveness Sleeve: a separate, pre-funded `tabarru` pool — deliberately distinct from the `InsuranceFund` of §5.2.1, and junior to all `takaful` claims, so policyholder backing and validator payroll never compete in the same crisis — governed by five rules:

1. Counter-cyclical fill. The sleeve skims 10% of transaction fees in epochs where volume exceeds its long-run median, stopping at a cap of 12 months of aggregate reference OpEx (a published, board-ratified cost benchmark — not validator self-report). A genesis allocation seeds roughly a third of the cap so the chain is never unprotected while the skim accumulates (the skim fills the remainder in ≈ 3 –4 normal years).
2. Conjunction trigger, both legs sustained. Activation requires volume below $0.35 \times$ normal for 24 consecutive hourly epochs and the published κ above its p99 governance constant for 24 consecutive epochs.

Volume alone never suffices: measured volume is the one signal validators control (a cartel can censor transactions to suppress it), so it cannot be the sole key to the vault. Thresholds are fixed governance constants recalibrated annually by ordinary governance — trailing-window quantiles would themselves be gameable by slow grinding.

3. Shortfall payout. When active, the sleeve pays each validator $\max(0, \beta \cdot \text{OpEx_ref} - \text{fee income})$, performance-weighted by consensus-measurable work during the stress window (blocks signed, vote-extension participation, κ submissions within tolerance of the median, uptime) and decaying 2% per week in subsidy. The replacement ratio $\beta = 1.3$ is a *dispersion allowance*, not a profit margin: OpEx_ref is an aggregate benchmark, and the upper tail of the audited validator cost distribution needs $\approx 1.2\text{--}1.4\times$ the reference to reach its own floor.
4. Hysteresis and exit. Deactivation requires volume above $1.3\times$ the trigger floor sustained for one week, followed by a one-month cooldown before any re-trigger.
5. Governance separation. Trigger and payout parameters are DAO-tunable within board-bounded invariants (as with the 60/20/20 split); the Shariah board bounds the mechanism but does not operate the trigger, since its own treasury is fee-funded.

Why it is incentive-compatible. The defense against gaming is the payout formula, not the trigger. In Monte-Carlo experiments (six-year hourly paths, thirty heterogeneous validators, joint volume-and- κ crises calibrated to the companion papers' estimated κ dynamics), a 35%-stake cartel suppressing half of measured volume *can* force the trigger on a majority of paths — and loses money on every one of them (the attack's net payoff is strictly negative; best case across all simulated paths remains deeply negative). The shortfall structure means a validator must destroy its own fee income below the floor to unlock a payment smaller than what it destroyed, and subsidized income runs at 71–77% of normal income, so no validator prefers the subsidized state. The trigger conjunction showed a 0% false-positive rate on paths without a true winter: minor downturns never reach the sleeve, so there is no “lazy validator” equilibrium in ordinary volatility.

What it buys. In the same simulations, validator survival through a severe winter (volume -75% , half-life nine months) rises from 84% to 90%, and through a black-swan winter (-85% , half-life twelve months) from 67% to 78% — with the pool's black-swan minimum approaching zero, which is precisely why the cap is set at twelve months rather than the six a moderate-winter analysis would suggest.

Why it is not riba through the back door. A badly parameterized sleeve — generous, non-decaying, stake-weighted rather than performance-weighted — would be a predetermined return for the passage of time, paid regardless of work: the inflationary-emission structure this chain abolished, reintroduced under the label of “security.” The parameters above are what keep the payment *tabarru*: outcome-contingent (only on measured shortfall), earned (performance-weighted, only for work done during certified stress), bounded (capped at audited cost, decaying), and exceptional (conjunction-triggered, board-certified). The anti-moral-hazard conditions and the anti-riba conditions are the same conditions — returns must track real work and real risk, at the consensus layer as everywhere else in the system.

Status: design with simulation evidence (reference model and parameter grids in `sleeve_model/`); not yet implemented in the reference modules. The volume process in the simulations is a scenario calibration — no chain history exists yet — and the model's limitations are documented alongside it.

5.3 EVM application layer

The κ -Chain uses the Cosmos native EVM framework (v0.5.0, security-audited release) for its execution layer. The Cosmos native EVM is maintained by the Cosmos core team, is IBC-native by construction, and includes EIP-7702 account abstraction — enabling simplified user interactions without requiring users to manage separate gas wallets. Production deployments as of 2026 include Ripple, MANTRA, Mezo, and KiiChain, confirming enterprise-grade stability.

The EVM layer shares global state with the native modules: - Smart contracts can read `block.kappa` as a native opcode - Smart contracts can call `x/perp`, `x/credit`, and `x/takafu` via precompile addresses - Gas fees paid in BRKX (the native token) - Solidity/Vyper compatibility: existing Baraka Protocol contracts deploy with minimal modification - Account abstraction (EIP-7702): users interact without managing BRKX gas manually — onboarding for Islamic finance users unfamiliar with crypto gas mechanics

Baraka Protocol v3 (currently on Arbitrum Sepolia) is the reference implementation for the EVM layer. The migration path from Arbitrum to κ -Chain involves substituting block.kappa for the Chainlink oracle feed and removing the ι enforcement guard (which becomes unnecessary as the substrate guarantees $\iota = 0$). Because Cosmos native EVM maintains an IBC channel from genesis, Phase 1 Arbitrum users can bridge positions to the κ -Chain via IBC v2 rather than a custom bridge — a message-passing migration, not a liquidity migration event.

5.4 IBC interoperability

As a Cosmos SDK chain, the κ -Chain natively supports the Inter-Blockchain Communication protocol. The Cosmos 2026 roadmap delivers IBC v2 with light client support for Solana and a general solution for all EVM and L2 chains — meaning the κ -Chain connects to Ethereum, Arbitrum, Solana, and every major L2 via a single protocol without custom bridges.

Planned IBC connections: - Arbitrum (Phase 1 → Phase 3 migration): Baraka Protocol positions on Arbitrum are IBC-portable to the κ -Chain via IBC v2 — a message-passing migration, not a liquidity event - USDC / Noble: stablecoin collateral bridge from Ethereum for x/perp margin and x/credit settlement - Osmosis DEX: BRKX tradeable on Osmosis, the largest Cosmos DEX - HAQQ Network: direct channel to the Islamic L1; BRKX airdrop distribution to ISLM holders - QFC HashSphere / regulatory on-ramp: Qatar Financial Centre's permissioned Hedera DLT for Islamic finance compliance receipts; κ -Chain sukuk issued on-chain, compliance records on QFC DRS — combining open market pricing with regulated institutional record-keeping - OIC central bank chains: Bank Negara Malaysia and SAMA CBDC programmes are both Cosmos-compatible; IBC channels are available once those chains launch

IBC v2's cross-chain light clients mean any participant on any of 100+ connected chains can access κ -Chain instruments without running a separate wallet or using a bridge. The network effects of the Cosmos ecosystem become the κ -Chain's distribution layer.

5.5 Validator set design

Phase 3a: ICS consumer chain launch (cold-start solution) The hardest problem for any new Cosmos appchain is validator bootstrapping. No Islamic financial institution has ever run a blockchain validator node. Requiring IsDB, BNM, and ADIB to acquire and stake BRKX before genesis — with no track record, no liquidity, and no operational infrastructure — is the hardest possible version of this problem.

The solution is Cosmos Interchain Security (ICS): the κ -Chain launches as a consumer chain on the Cosmos Hub, borrowing its 180+ validator set for initial security. This requires zero validator recruitment, zero BRKX staking at genesis, and provides enterprise-grade Byzantine fault tolerance from day one. Consumer chains maintain full sovereignty over their own token, fee model, and governance — ICS only shares the validator set, not the economic model.

The κ -Chain's wakala fee structure (fees from real activity, no inflationary emission) is fully compatible with ICS: the ICS protocol does not impact the transaction fee system on consumer chains. Cosmos Hub validators earn their share of κ -Chain fees in addition to their ATOM staking rewards.

Phase 3b: Proof-of-Authority institutional onboarding (parallel track) While running on ICS, the κ -Chain uses the Cosmos 2026 Proof-of-Authority (PoA) module — which requires no Cosmos SDK fork and no stake token — to begin onboarding Islamic financial institutions as validators by institutional identity rather than by capital stake.

PoA admission criteria: - Shariah board approval as a Shariah-compliant institution - Signed Validator Operating Agreement ($\iota = 0$ enforcement, public identity disclosure) - Demonstrated operational capability (node uptime test on κ -Chain testnet for ≥ 90 days) - Minimum institutional AUM of \$1B in Islamic finance assets

An important refinement borrowed from the IsDB's Proof-of-Use (PoU) patent (US 12,548,031 B2, granted 10 February 2026, IsDB Institute): validator governance weight within x/gov should be tied to actual usage of the chain's financial instruments, not solely to stake size. An institution that actively issues sukuk, contributes to takaful pools, and facilitates iCDS on the κ -Chain should have proportionally greater governance influence than

one that merely holds the minimum stake and sits idle. This aligns governance incentives with real economic participation — the PoU principle applied to the κ -Chain’s institutional validator set.

Target institutional validators: - Islamic Development Bank (IsDB) — Jeddah (*PoU patent holder; priority engagement*) - Bank Negara Malaysia / Khazanah — Kuala Lumpur - Abu Dhabi Islamic Bank (ADIB) — Abu Dhabi - HSBC Amanah — Dubai/London - Bank Islam Malaysia — Kuala Lumpur - Saudi National Bank Islamic division — Riyadh - Qatar Islamic Bank — Doha (*QFC Digital Asset Lab participant*) - Bank Syariah Indonesia (BSI) — Jakarta - Kuwait Finance House — Kuwait City - Amundi Islamic — Paris

Phase 3c: Sovereign PoS migration Once 5+ institutional validators are running on PoA with 6+ months of operational stability, and BRKX has sufficient market liquidity for institutional acquisition, the κ -Chain migrates from ICS to sovereign proof-of-stake. Validators transition from PoA identity-based admission to PoS stake-based security, with Shariah board approval as an additional admission gate above the economic threshold.

Validator economics: With \$500M annual notional volume at 5bps, total protocol fees are \$250K/year; at the 60% proposer share across 20 validators that is $\approx$ \$7.5K/validator/year — illustrative, and frankly below sustainability at that volume. Validator income scales linearly with throughput: the model only becomes Cosmos-Hub-comparable at materially higher volume (e.g. \$500M daily notional $\rightarrow$ $\approx$ \$2.7M/validator/year). What matters structurally is that *all* of it derives from real fee activity — the κ -Chain has no inflationary block-reward emission, so there is no dilution risk at any volume.

6. The κ -yield curve as monetary policy instrument

6.1 The current dual-system problem

Central banks in GCC and Southeast Asia currently operate dual monetary systems. Bank Negara Malaysia publishes both a conventional overnight rate and an Islamic overnight rate (KLIBOR vs IIMM rate). The Saudi Arabian Monetary Authority maintains similar parallel systems. These Islamic rates are administered by decree — the central bank sets them. They are not market-determined.

The consequence is that Islamic monetary policy lacks the open-market operations mechanism available to conventional central banks. A conventional central bank can target rates by buying and selling bonds in the secondary market — the market rate moves toward the target through arbitrage. An Islamic central bank has no equivalent secondary market with a genuine *riba*-free rate to target.

The κ -yield curve, as estimated in Ahmed & Bhuyan (2026c), provides the first empirically-grounded, market-determined, non-negative Islamic rate term structure. It is derived from secondary sukuk market activity, not administered by any authority. It is *riba*-free by construction — there is no discount rate in the formula, only the hazard intensity κ ; product-layer Shariah questions remain with scholars.

6.2 On-chain monetary policy mechanics

When sovereign sukuk are issued and traded on the κ -Chain, the secondary market prices update the κ -yield-curve NSS state (validator-fitted off-chain, aggregated by stake-weighted median in PreBlocker — §5.1), which reprices x/credit and $x/\text{takaful}$ instruments; the x/perp per-market κ values remain computed from each perpetual’s own basis (§6.3). The chain publishes κ -yield curve parameters as consensus-validated state every block (read via the `block.kappa` precompile). A central bank operating on the κ -Chain could:

1. Issue sovereign sukuk on-chain at a target κ , priced using the current κ -yield curve
2. Conduct open-market operations by buying or selling on-chain sukuk in the x/credit secondary market, shifting the term structure of κ
3. Observe the transmission in real time: the published κ -yield curve shifts with the OMO, repricing on-chain sukuk and credit instruments at the next block; any pass-through to the perpetuals basis runs through traders’ expectations, not a mechanical link

This is Islamic QE. Not as an aspiration — as a mechanically specified protocol.

The Feller condition guarantee ($2\alpha\bar{\kappa} \geq \nu^2$) means κ can never go negative under CIR dynamics. This is a structural property that conventional central banks, operating with r as their policy rate, cannot guarantee — as the ECB, Bank of Japan, and Swiss National Bank discovered between 2014 and 2022 when their policy rates went below zero and their Islamic finance instruments became technically mispriced.

6.3 The κ oracle as public good

The κ -Chain's publication of κ -yield curve parameters as consensus-validated state every block is a public good for the entire Islamic finance industry. Any institution — on-chain or off-chain — can read the chain and price their instruments against genuine market-determined *riba*-free rates.

Two clarifications are necessary. First, the chain publishes per-market κ values — κ_{BTC} , κ_{ETH} , and any other listed perpetuals asset — each computed independently from its own basis. These are distinct numbers. The chain does not aggregate them into a single κ . An institution building a *sukuk* uses its own instrument-specific κ (from §2.5.2 Source 1 or Source 2), which it can calibrate against the chain's published κ -yield curve parameters as a consistency check.

Second, this benchmark is the Islamic equivalent of LIBOR/SOFR — but with four structural improvements: it is non-negative by construction (Feller condition), computed from a genuine secondary market rather than administered by a panel bank survey, *riba*-free at the pricing layer by construction (certification of products built on it rests with Shariah boards), and plural by design rather than a single rate that creates systemic concentration risk. The 2012 LIBOR manipulation scandal is impossible in a system where each κ is computed independently from each market's own on-chain trading activity.

7. The political economy case

$\iota = 0$ is not a modern regulatory patch on fractional-reserve banking; it operationalizes a monetary ontology articulated in the classical Islamic canon. Al-Ghazālī (*Iḥyā' 'Ulūm al-Dīn*) defined money as a mirror — a thing with no colour of its own that exists only to reflect the value of real goods — and held that lending money to breed money forces the mirror to generate its own colour, subverting its purpose. Ibn Taymiyyah cast money as an immutable *mī'yār* (standard of value) whose integrity the authority must protect; Ibn Khaldūn tied legitimate return to human labor and transaction velocity rather than idle capital. The κ -Chain's substrate invariants — $\iota = 0$ enforced in consensus, a manipulation-resistant κ metric, and a fee-only (non-inflationary) validator reward — are mechanical instantiations of these principles, not retrofits. The full treatment, with sources, is in the companion paper *Classical Foundations of the κ -Substrate*; this section develops the modern political-economy consequences.

7.1 Illiquidity as the mechanism of extraction

The Spring-Loaded Door computational political economy project (MASTER_FINDINGS.md, companion project) models democratic backsliding as a function of the L8 wealth-to-power converter: concentrated interests organise and capture institutions while diffuse interests cannot. The v13 composite apparatus classifies 16/16 historical backsliding cases binary-correctly and predicts trajectories in 7/10 cases within ± 0.10 libdem.

The framework's most important substantive finding for this white paper is §0.5.4: illiquidity is the bargaining position. The garment worker without income redundancy cannot refuse below-subsistence wages. The rice farmer who must sell at harvest cannot wait for a better price. These are not separate phenomena — they are the same structural mechanism operating at two scales of the L2 distribution layer.

The v6 economic model identifies L4 monetary plumbing as the intervention layer: the place where payment rails and liquidity access directly change the bargaining position of forced sellers. Reform at L4 can interrupt the L8 wealth-pump without requiring legal change (L7), cultural shift (L6), or state capacity development (L3) — all of which are slower and more fragile.

The Bangladesh producer liquidity application (companion paper, `Bangladesh_Producer_Liquidity.md`) traces the mechanism end-to-end: 1943 cabinet decisions in London → 3 million deaths + multi-generational cardiometabolic programming (Paper 4 biological reparations) → structural illiquidity of rice farmers in 2026 → the April 2026 Labour Bill and EU CSDDD supply-chain due diligence regime as the current juncture window.

7.2 The κ -Chain as L4 infrastructure

The κ -Chain is, in the v6 framework’s terminology, an L4 intervention. Its instruments provide:

For rice farmers (Bangladesh, Pakistan, Indonesia, India): Actuarially fair crop insurance at $\pi^* = \hat{\kappa} \cdot B$, where $\hat{\kappa} \in [0.071, 0.147]$ depending on country and crop type. For a Bangladesh farmer with \$500 coverage, the annual premium is approximately \$35–74 per year. The TakafulPool smart contract implementation costs \$0.14 per transaction. The BADC warehouse receipt network provides the delivery infrastructure.

For garment workers: Micro-sukuk savings instruments structured as risk-sharing certificates: expected *ijarah/mudarabah* profit distributions (historically $\approx 2\text{--}4\%$ p.a.) that are variable, asset-linked, and loss-bearing — not a guaranteed deposit yield, which would reconstitute *riba*. For a garment worker earning \$200/month, such income redundancy directly increases their bargaining position.

For sovereign borrowers in OIC states: Sukuk issuance priced in κ rather than SOFR, removing the implicit dependence on a conventional benchmark rate at the pricing layer (Ahmed & Bhuyan 2026c) — the SOFR contamination that currently runs through Islamic sovereign balance sheets even when the instrument’s legal form is Shariah-compliant.

These are not development-aid transfers. They are correctly-priced financial instruments made available at the cost κ correctly implies, rather than at the higher cost that *riba* loading imposes. The difference is the *riba* loading itself: for agricultural takaful, 7.4% of premium at $r = 8\%$; for the iCDS, the present-value discounting haircut on the future loss (under 4% of notional); for sovereign sukuk, the implicit conventional-benchmark dependence the κ -curve removes.

7.3 The τ_{eff} implication

The Spring-Loaded Door framework shows that reform sustainability depends on $\tau_{\text{eff}} = f(C, \rho, \text{EMP})$: the decay timescale of reform is longer when state capacity C is high, cultural cohesion ρ is high, and L8 status EMP is low. The v13 apparatus predicts that 66% of states globally are currently in firing-classified mode — conditions where capture attempts are structurally possible.

The κ -Chain changes the τ_{eff} calculation for participating states in two ways:

First, a sovereign sukuk market priced in κ , rather than SOFR, reduces the dependence of Islamic sovereign borrowing on conventional financial system infrastructure. When a state can issue debt without implicit *riba* — and when its central bank can conduct monetary policy using the κ -yield curve — it reduces the leverage that conventional financial system actors hold over its policy choices.

Second, agricultural takaful and working-capital murabaha that reach smallholder farmers and informal workers directly build the income redundancy that is the individual-level mechanism through which L4 change translates to L2 bargaining power. The political-economic model names this as the handle. The κ -Chain provides the infrastructure for operating it.

8. Token economics

8.1 BRKX as L1 native token

On the κ -Chain, BRKX transitions from an ERC-20 governance token on Arbitrum to the native gas and security token of a sovereign blockchain. Its properties:

Supply: 100,000,000 BRKX total supply. Fixed. No mint function. Supply can only decrease through burn.

Functions: - Gas payment for all EVM-layer transactions - Validator staking (minimum 1,000,000 BRKX per validator) - Governance voting in x/gov Track 1 (DAO track) - Fee discount tiers for x/perp trading (inherited from Baraka Protocol design)

Value basis: BRKX is backed by the fee income flowing through the chain — perpetuals trading fees, sukuk issuance fees, takaful premiums, iCDS spreads. Unlike most Layer 1 tokens, BRKX has no inflationary emission. Its value is supported entirely by real economic activity denominated in κ -adjusted units. This is structurally analogous to a commodity-backed currency — but the commodity is the convergence intensity of a live Islamic derivatives market.

Riba-free by construction: BRKX staking earns from protocol fees (legitimate wakala-style agency income), not from an inflationary emission that dilutes non-stakers (which would be structurally similar to riba).

8.2 Launch mechanics

Phase 1 (Arbitrum, current): BRKX as ERC-20, governance token for Baraka Protocol.

Phase 2 (Cosmos native EVM): as the κ -Chain stands up, BRKX migrates from the Arbitrum Sepolia ERC-20 to the chain's native x/evm module, earning real trading-fee income from Baraka running on infrastructure the protocol controls — never a third-party $r > 0$ settlement layer.

Phase 3 (κ -Chain genesis): BRKX migrated to native token status. Existing BRKX holders receive the native token at 1:1. Validators lock BRKX as stake at genesis. The Shariah board publishes the founding fatwa for the chain itself (distinct from Baraka Protocol's fatwa, which covers a single DEX application).

Airdrop strategy (Hyperliquid model applied to Islamic finance): 310,000,000 HYPE was airdropped to early Hyperliquid users, creating an aligned user base that became the protocol's most powerful marketing force. BRKX community airdrop targets: - Early Baraka Protocol testnet users (technical validators of the mechanism) - GCC Islamic finance community members (via partner Islamic fintech apps) - Southeast Asian Islamic banking customers (via Bank Islam Malaysia, BSI partnership programmes) - Academic and scholarly community with established Islamic finance credibility

9. Governance

9.1 Dual-track architecture

The κ -Chain's governance inherits and extends the Baraka Protocol GovernanceModule design to the chain level.

Track 1 — DAO (community governance): Token holders create proposals, vote with BRKX, and queue approved proposals for execution after a 48-hour timelock. Standard community governance for fee parameters, new market listings, chain upgrade proposals.

Track 2 — Shariah board (compliance governance): The Shariah Advisory Board holds a 72-hour veto window after any proposal is queued. Their veto is irrevocable. No DAO vote, however large, can override the Shariah board on compliance matters.

The board's exclusive authorities: - Approval or delisting of assets from the trading whitelist - Approval of collateral tokens - Maximum leverage per market (5x genesis default; board-adjustable downward only, never above the 10x hard ceiling) - ι parameter (permanently 0; the board's mandate prohibits them from approving $\iota > 0$) - Fatwa publication and update (3-of-5 quorum, IPFS CID on-chain) - Emergency halt of any module without DAO vote

The immutability guarantee: The combination of these two tracks means that $\iota = 0$ cannot be changed by any governance action short of a coordinated simultaneous breach by both the DAO supermajority and the Shariah board. The Shariah board, whose mandate is defined by the founding fatwa, cannot vote for $\iota > 0$ without violating that mandate and losing its authority. This is the political-economic equivalent of embedding $\iota = 0$ not just in code but in the founding constitutional document of the chain.

9.2 Shariah board composition and mandate

The founding board consists of five AAOIFI-recognised scholars with expertise across Islamic banking law (fiqh al-muamalat), contemporary finance, and smart contract technology. The board:

- Meets quarterly in person and on-chain via ComplianceOracle
- Publishes fatwa documents (IPFS, on-chain CID) for every new instrument category
- Responds to community-submitted Shariah questions within 30 days
- Receives annual compensation from the Shariah Board operational treasury (20% of protocol transaction fees per the wakala split of §5.2.1 — there are no block-reward emissions)

Board decisions require 3-of-5 quorum. The two-step transfer rule (board can only change composition if incoming member actively accepts the role) prevents hostile replacement.

10. Implementation roadmap

Phase 1: Prove the mechanism (Q3 2026 — seed round \$400K)

Deliverables: - Baraka Protocol v3 on Arbitrum mainnet: 13 contracts live, external security audit (Code4rena or equivalent), fatwa from AAOIFI-recognised scholars - κ oracle v1: computing convergence intensity from live perpetuals basis on Arbitrum, published to a public endpoint - IFA26 conference presentation (June 2026, Kuala Lumpur): academic validation of the $\iota = 0$ mechanism and κ -rate framework in front of the Islamic finance scholarly community - Paper 1 (SSRN 6322778) accepted by journal: peer-reviewed academic validation of the core claim

Success criteria: Real trading volume on Baraka mainnet; κ oracle producing stable readings; at least one AAOIFI scholar willing to engage with the κ -rate framework.

Phase 2: Distribute and anchor the rate (Series A ~\$3–5M, 12–18 months)

Deliverables: - First listings: asset-linked perpetuals satisfying R1–R4 — κ -Gold (vault-secured tokenized gold; inflation and currency hedging for businesses in volatile-currency jurisdictions) and κ -Sukuk-yield (institutional balance-sheet liquidity management replacing interest-rate swaps) — the risk-transfer instruments that lead the launch while the cash-settled perpetual remains subject to the open SBR-5 fatwa - Baraka Protocol on Arbitrum mainnet: riba-free ($\iota = 0$) perpetuals DEX with real liquidity and a live κ oracle — Phase-1 hardening ahead of the native Cosmos EVM (x/evm) migration (the perpetuals remain subject to the open SBR-5 fatwa on gharar/maysir; see §11) - κ -yield curve published on-chain from live sovereign sukuk data, updated weekly - Pilot sovereign sukuk issuance: working with one GCC or Malaysian sovereign to issue a small sukuk tranchised priced in κ , not SOFR - Papers 2–5 accepted: academic validation of iCDS, takaful, and κ -yield curve findings - Bank Negara Malaysia or SAMA exploratory engagement: presenting κ -yield curve as potential Islamic monetary policy benchmark

Success criteria: \$50M+ notional volume on Baraka; one sovereign sukuk priced in κ ; two papers accepted in peer-reviewed journals; one OIC central bank expressing interest in the κ benchmark.

Phase 3: Sovereign Islamic L1 (Series B ~\$15–30M, 3–5 years out)

Phase 3a — ICS consumer chain launch: - Cosmos SDK κ -Chain: Go implementation of x/perp, x/shariah, x/credit, x/takaful, x/gov modules - Launch as ICS consumer chain on Cosmos Hub — inherits 180+ validators, no cold-start problem - BRKX migration to native L1 token; ICS fee sharing with Cosmos Hub validators - IBC v2 connections: Arbitrum (position migration), HAQQ, Osmosis, Noble (USDC), QFC HashSphere

Phase 3b — Institutional validator onboarding (parallel): - Cosmos PoA module: IsDB, BNM, ADIB onboarded as identity-based validators, no BRKX stake required at entry - IsDB engagement on PoU patent integration: usage-weighted governance in x/gov - QFC Digital Asset Lab partnership: κ -Chain sukuk issued on-chain, regulatory

receipts on QFC DRS - Agricultural takaful live pool: Bangladesh or Indonesia rice farmers, parametric trigger via World Bank yield index

Phase 3c — Sovereign PoS migration (5+ PoA validators stable, 6+ months): - Transition from ICS to sovereign PoS; institutional validators add BRKX stake - κ -Chain fully independent: own security, own validator set, own monetary policy - Academic white paper: this document, peer-reviewed and published

Success criteria: \$1B+ TVL; one central bank using κ -yield curve in monetary policy analysis; 500,000+ monthly active users; agricultural takaful pool with 10,000+ farmer participants; IsDB and at least two other IFIs running as sovereign validators.

11. Honest limitations and open questions

11.1 What is proven

The mathematical results in Papers 1–5 are proven in the sense that academic mathematical proofs are proven: the Separation Theorem follows from the AHJ no-arbitrage result and the Duffie-Singleton isomorphism by standard mathematical argument. The κ -yield curve empirical results have been computed on real market data and are reproducible from publicly available sources. The agricultural takaful comparison against PMFBY is an empirical observation — a units-level agreement, not a strict validation (§3.3).

The simulation results (Paper 6, cadCAD + RL + game theory + mechanism design) provide computational validation that the protocol operates as claimed at $\iota = 0$ under adversarial trader conditions. Zero insolvency events across all 30 Monte Carlo seeds (5 episodes $\times$ 720 steps each); Nash equilibrium leverage of $2.72 \times / 3.28 \times$ in the headline seed and $3.16 \times / 2.84 \times$ across $R = 30$ seeds (well below the $5 \times$ genesis leverage cap); mean net transfer of $-\$660$ single-seed and $+\$865$ (95% CI $\$245$ – $\$1,485$) across $R = 30$ — $\leq 0.9\%$ of notional, consistent with no systematic riba-like transfer.

11.2 What is not yet proven

The Shariah ruling on perpetual futures. The foundational SBR-5 question — whether perpetual futures contracts are a valid Islamic financial instrument even at $\iota = 0$ — has not yet received a scholar-signed fatwa. This is the single most important open question. The mathematical proof that $\iota = 0$ eliminates riba is necessary but not sufficient; scholars must also rule on gharar (excessive uncertainty) and maysir (gambling) aspects of the contract structure. The SeekersGuidance fatwa (October 2025) is encouraging but not authoritative. AAOIFI engagement is the required next step.

Crucially, this open question is confined to *cash-settled synthetic* instruments. Riba is resolved at the substrate ($\iota = 0$), but gharar (asset existence and deliverability), qabd (possession), and maysir (intent-based speculation) are product-layer questions, and the asset-backed instrument families address them directly: sovereign sukuk and commodity structures carry a real, deliverable underlying — resolving existential gharar and enabling qabd (constructive possession of a registered asset, with QFC Digital Receipt System records and, for agricultural takaful, the BADC warehouse-receipt delivery rail) — and parametric takaful is a *tabarru* (mutual-donation) structure tied to real loss rather than a zero-sum price bet, which structurally avoids maysir. The riba-free substrate is therefore a *necessary foundation* on which Shariah-compliant products are built under board guidance — not a blanket compliance claim for every instrument, least of all the cash-settled perpetuals, which remain the primary subject of the SBR-5 fatwa. (The classical jurisprudence on these three pillars — al-Dharir on gharar, AAOIFI Standard 20 and Kamali on qabd and futures, Bacha and Obaidullah on maysir — is reviewed in the companion Shariah-boundaries note.)

The x/shariah module as a legal compliance layer. On-chain Shariah compliance is new. There is no established jurisprudential precedent for a smart contract module that enforces compliance rules at the transaction level. The AAOIFI standard-setting process has not yet addressed this. Our view is that on-chain enforcement is more rigorous than conventional Shariah auditing, which is periodic and retrospective — but this view has not been validated by AAOIFI.

The Cosmos SDK appchain. Phase 3 requires rewriting the Baraka Protocol core from Solidity to Go. This is a substantial engineering undertaking. The AI-assisted timeline estimate is 12–18 months from a Series A close with a dedicated engineering team. The estimate is informed by the modularity of the Cosmos SDK and the availability of dYdX v4’s open-source code as a reference implementation, but it is an estimate, not a guarantee.

Regulatory status. The κ -Chain, as a sovereign Layer 1 blockchain, will require legal entity formation in at least one jurisdiction with a clear crypto-asset regulatory framework. VARA (UAE) and the Bahrain CBB are identified in Ahmed & Bhuyan (2026e) as the most permissive pilot frameworks for the iCDS. The chain-level regulatory analysis has not yet been conducted.

The κ oracle stability and the thin-market fallback. The on-chain κ oracle computes convergence intensity from the perpetuals basis. In low-volume periods — guaranteed in the early phase of any new protocol — the basis is thin and the perp leg becomes cheap to push, so κ must fall back to the κ -yield curve estimated from external sukuk data via a Shariah-board-approved oracle. The design of that handover is itself a security surface. A naive hard switch at a volume threshold is *manipulable by whipsaw*: an adversary oscillating volume across the threshold flickers the published κ between the thin perp leg and the sukuk leg. A Monte-Carlo model of exactly this attack (Paper 3 §“Systemic and adversarial robustness,” `simulations/thin_market/`) shows the hard switch jumping 165 bp in a single block and tripping the margin-call trigger 13 times per run — each a potential liquidation cascade. The κ -Chain therefore uses a continuous gated blend, not a switch: $\kappa = w \cdot \kappa_{\text{perp}} + (1-w) \cdot \kappa_{\text{sukuk}}$, where the weight $w = \text{smoothstep}(\text{volume})$ runs on an asymmetric EMA (slow to trust new liquidity, fast to distrust a thinning book) and a dispersion gate freezes the perp leg once validator price-spread exceeds a band. The same model shows this holds the worst single-block move below the liquidation trigger with zero cliffs — a smooth, non-whipsawable handover. The discontinuity, not the κ level, was the hazard; a C^1 boundary removes it. The sukuk-curve leg remains a temporary external dependency that the Phase 2 milestones are designed to eliminate.

Cross-market contagion. Once the chain lists more than one κ -market (κ -Gold, κ -Sukuk, κ -Compute) against a shared collateral asset, a forced liquidation in one market depresses the collateral price that marks all of them — a classic contagion channel. A three-market cadCAD model (`simulations/contagion/`) shows the cascade is threshold-like: a small shock stays local, but past a critical size a naive configuration tips into a system-wide cascade in which the non-shocked books fully liquidate. The substrate’s existing defenses — the ± 75 bp per-epoch κ clamp (§5.2), a per-block liquidation throttle, and a takaful collateral floor — keep the same shock sub-critical, containing it to the originating market. The clamp and throttle are therefore not merely per-market circuit breakers but the chain’s cross-market firewall, and their calibration is a system-level safety parameter. (The same takaful machinery extends forward to a future κ -Compute *salam* market, whose provider-default backstop a sizing sweep funds at an 8% tabarru rate — `simulations/compute_salam/`.)

The IsDB Proof-of-Use patent. The Islamic Development Bank Institute holds US Patent No. 12,548,031 B2 (“*Method and system for transaction validation in a distributed computing system*,” granted 10 February 2026) for a consensus mechanism that ties validation authority to actual network usage rather than capital stake. The κ -Chain uses CometBFT, which does not infringe. But the patent raises a strategic question: should the κ -Chain integrate PoU principles into its validator governance layer — specifically, weighting governance influence by protocol usage rather than stake size alone? This requires formal engagement with IsDB’s patent team before implementation. The engagement is also the most direct pathway to IsDB validator participation, making it doubly urgent.

The QFC HashSphere regulatory question. Qatar Financial Centre’s HashSphere is a permissioned Hedera-based DLT for Islamic finance compliance records. An IBC connection between the κ -Chain and QFC’s infrastructure would make the κ -Chain the first public blockchain with a QFC-regulated institutional on-ramp. But the regulatory treatment of an IBC connection to a public chain from a permissioned institutional ledger has not been tested. QFC may require the κ -Chain to meet specific KYC/AML requirements at the channel level before permitting the connection. This is an open legal and regulatory question requiring QFC Digital Asset Lab engagement in Phase 2.

11.3 The bootstrap problem

Every new blockchain faces a bootstrap problem: validators need users, users need applications, applications need liquidity, liquidity needs validators. The κ -Chain's bootstrap path is:

Phase 1 (Baraka on Arbitrum) generates real users and real κ data without requiring the κ -Chain to exist yet. By the time Phase 3 begins, there is a documented track record of $\iota = 0$ operating in production, a live κ oracle with two years of data, peer-reviewed academic papers validating the mechanism, and a community of users who understand the value proposition.

This is the same path dYdX took: Ethereum contracts (2017–2023) before the appchain (2023). The difference is that Baraka Protocol is already deployed.

12. Related work

dYdX v4: The architectural inspiration for the κ -Chain's on-chain CLOB design. dYdX v4 operates at $\iota = 0$ by default but as a technical choice, not a commitment — governance vote 220 introduced $\iota = 0.125\text{bps/h}$ on isolated markets. The κ -Chain makes $\iota = 0$ immutable. dYdX also demonstrated the correct sequencing: Ethereum contracts first, appchain second, after user base and volume justify the migration cost. Baraka Protocol follows the same path.

HAQQ Network: The most instructive comparison for the κ -Chain's positioning. HAQQ has \$400M in funding and a Shariah Oracle that verifies smart contract compliance before deployment — among the best-funded Islamic L1s in existence. But HAQQ's block reward model issues newly minted ISLM at a predetermined announced rate to validators and stakers. This is structurally *riba* at the substrate: a fixed predetermined return for the passage of time, beneath the Shariah Oracle's visibility. The Oracle checks applications; the inflationary emission is in the consensus layer. HAQQ is the living proof that \$400M and scholar endorsement cannot solve the monetary primitive problem without addressing it at the substrate. The κ -Chain's fee-only wakala block reward (no inflation, no predetermined rate, earned from real activity) resolves what HAQQ does not. Strategic relationship: complementary, not competitive — IBC connection, BRXX airdrop to ISLM holders. What to borrow: HAQQ's Evergreen DAO model (10% of protocol revenue to an Islamic public goods fund, scholar-endorsed, community-governed) is directly applicable to the κ -Chain's Shariah Board treasury design.

Setheum (SetMint): A Substrate-based Islamic L1 (EVM-compatible, forkless) whose flagship primitive is SetMint — a zero-interest, zero-stability-fee over-collateralised CDP that mints the SETUSD stablecoin, alongside Musharakah-based spot pools. It is the closest *sovereign-L1* analogue to the κ -Chain in intent, which makes the contrast sharp. First, scope: Setheum targets spot stablecoins and lending — the mathematically solved part of Islamic DeFi — whereas the κ -Chain targets perpetuals and the *riba-free pricing primitive*. Second, and decisively, Setheum inherits the standard Substrate/PoS substrate, whose inflationary staking emission is itself a predetermined return for the passage of time (*riba al-nasi'ah* at the consensus layer) — the same wrong-layer problem as HAQQ, which the κ -Chain's fee-only wakala reward resolves. Setheum appears to remain pre-mainnet at the time of writing; this is therefore a comparison of designs, not of live deployments. (Note: its mechanism is "SetMint," not the "ECDP" label sometimes attached to it in secondary commentary.)

Fasset ("Own" L2): An *infrastructure* competitor — and the most institutionally advanced one — not a DApp. Fasset holds a 2025 Malaysian provisional licence for a stablecoin-powered Islamic digital bank, raised \$51M in 2026 (SBI, Investcorp, Arz Portföy), reports multi-billion-dollar annualised volume, and is building "Own," its own Ethereum L2 on Arbitrum to settle tokenised real-world assets. It is therefore building an alternative *rail* for regulated Islamic RWA, overlapping the κ -Chain's institutional ambition. The κ -Chain's differentiation is not distribution or licensing — where Fasset currently leads — but the monetary primitive: "Own," like every conventional L2, settles on an $r > 0$ substrate and prices RWA against conventional benchmarks, whereas the κ -Chain encodes $\iota = 0$ natively and publishes κ as a consensus-validated state primitive (the block.kappa precompile). The contest is Fasset's regulatory/distribution head-start versus the κ -Chain's substrate-level pricing purity.

Caiz and Blossom (adjacent, not competitive): Caiz (Caizcoin) is a fatwa-certified Islamic chain enforcing network-level KYC/AML through a hybrid ("DeCe") Islamic Federated Byzantine Agreement structure rather

than a public EVM substrate. Blossom Finance operates the pioneering SmartSukuk platform (Ethereum smart contracts, running in Indonesia since 2017). Neither produces a market-determined riba-free *rate*; both sit alongside, rather than against, the κ -Chain's pricing layer.

IsDB Proof-of-Use (PoU): US Patent No. 12,548,031 B2 (*“Method and system for transaction validation in a distributed computing system”*), granted 10 February 2026 — IsDBI's second USPTO patent (fifth overall). The Islamic Development Bank Institute's mechanism ties validation authority to actual network usage rather than capital stake — participants validate others' transactions in exchange for having their own validated. No IP conflict with CometBFT. Strategic significance: IsDB is the most important target validator and an active blockchain innovator whose consensus design philosophy directly addresses the PoS wealth concentration problem the κ -Chain's wakala fee model partially resolves. The PoU insight — usage-weighted governance influence rather than stake-weighted — is integrated into the κ -Chain's §5.5 validator governance design. Engagement: IsDB as founding institutional validator, PoU integration into x/gov as a co-developed feature.

Qatar QFC HashSphere: A private permissioned Hedera-based DLT launched September 2025 under Qatar Financial Centre's Digital Asset Lab (AlRayan Bank, Blade Labs, Hashgraph). Its Digital Receipt System provides transparency, compliance records, and QFC regulatory cover for Shariah-compliant asset-backed finance. HashSphere is compliance infrastructure — it tracks records of transactions priced elsewhere, using rates derived from elsewhere. It cannot generate κ , produce a yield curve, or operate as an open market. The κ -Chain and QFC HashSphere are at different layers: the κ -Chain prices and settles; QFC DRS provides regulatory receipts and KYC cover. The correct model is partnership via IBC connection — making the κ -Chain the first public blockchain with a QFC-regulated institutional on-ramp. Qatar Islamic Bank (target κ -Chain validator) is a QFC participant, making this relationship doubly strategic.

Hyperliquid: The performance benchmark and scaling model. Hyperliquid grew to \$9.5B OI and 1.4M users with 11 staff, no VC, no marketing — by building CEX-level performance and redistributing fees to users via the HYPE airdrop. The κ -Chain applies the identical playbook to the 1.8 billion people whose religious tradition prohibits using Hyperliquid. The κ -Chain uses Cosmos native EVM v0.5.0 — IBC-native, security-audited, and core-team maintained, on infrastructure the protocol controls. Hyperliquid's moat is performance. The κ -Chain's moat is permission. Performance can be copied. A mathematically-grounded $\iota = 0$ substrate cannot be forked without forking the mathematics.

Avalanche: Compliance as configuration — institutions run compliant subnets while benefiting from public blockchain infrastructure. The κ -Chain's position is compliance as proof: $\iota = 0$ is enforced at the consensus layer, backed by a mathematical theorem and a Shariah board veto, reversible by no governance action short of simultaneously breaching both the DAO supermajority and the board's founding mandate.

Circle Arc4: Circle's 2026 “Economic Operating System for the internet” validates the architectural thesis that specialised financial L1s are the correct design choice. Circle encoded the conventional financial system's axioms at the L1 level. The κ -Chain encodes a different axiom. Both are financial operating systems. They differ on the single most important parameter in financial mathematics.

Cosmos SDK 2026 roadmap: PoA module (institutional validator onboarding without BRKX stake), Cosmos native EVM v0.5.0 (security-audited, EIP-7702 account abstraction, IBC-native), and IBC v2 (light client support for all EVM chains and Solana) solve three of the five hardest Phase 3 technical problems without the κ -Chain team building them. The Cosmos ecosystem is the correct infrastructure choice.

Ethereum + DeFi: AAVE, Compound, and other lending protocols are $\iota > 0$ by design. They cannot be made Shariah-compliant by adding a governance parameter — the interest rate is embedded in their pricing model, and the Ethereum substrate pays inflationary staking yield. The κ -Chain does not try to make Ethereum DeFi halal. It provides an alternative substrate where the question does not arise because the primitive does not exist.

13. Conclusion

13.1 What the mathematics built

The κ -Chain rests on a precise mathematical foundation. The Separation Theorem (Ahmed & Bhuyan 2026b) establishes that interest is eliminable from no-arbitrage pricing. The κ -yield curve (Ahmed & Bhuyan 2026c) provides a non-negative, empirically grounded, market-determined rate — not a single rate, but a family of instrument-specific rates each calibrated from its own real economic data — that replaces r across the full instrument spectrum. The four closed-form pricing formulas (perpetuals at $f_t = E[X_\theta]$, risk-sharing sukuk at $P_{\text{sukuk}} = E[X_\theta]$, takaful at $\pi^* = \kappa B$, iCDS at $s^* = \kappa(1-\delta)$) are computable, on-chain verifiable, and riba-free by construction (product-layer Shariah questions remain with scholars).

κ is not a fancy word for interest. Interest measures the price of time — the riba component, the reward for waiting, charged regardless of economic outcome. κ measures the intensity of a real event: how fast a market converges, how frequently a crop fails, how likely a sovereign defaults. The same mathematical symbol appears in all four instruments because they are all stopping time problems. But they are calibrated independently, from independent data, measuring independent real-world phenomena. There is no central committee setting κ . The perpetuals market finds it continuously from the live basis. The takaful actuary calibrates it from meteorological records. The sukuk issuer derives it from project engineering timelines. A Bangladesh mortality table and a BTC futures basis both produce a κ . They share a mathematical language. They share nothing else.

This plurality is the proof that κ is not interest in disguise. Interest is one number, set by a committee, applied everywhere. κ is a surface — a map of real economic intensities across domains and maturities, each cell grounded in physical reality, each non-negative by the Feller condition. SOFR is a point. The κ -surface is a map of the world.

13.2 What the L1 architecture changes

Every existing blockchain encodes a foundational axiom in its consensus layer: *time has a price, and that price is $r > 0$* . This axiom is present in inflationary staking rewards, in fee assets priced using conventional discount rates, and in the complete absence of any native κ primitive in any existing state machine. It was never stated. It was never voted on. It is simply the default inherited from conventional financial mathematics, baked silently into the substrate of every chain that exists.

HAQQ has \$400 million in funding and a Shariah Oracle that verifies smart contracts before deployment. The Oracle checks applications. The riba is in the consensus layer, paying validators a predetermined return for the passage of time, beneath the Oracle's visibility. The best-funded Islamic blockchain in history could not solve the problem because it addressed the wrong layer.

The κ -Chain encodes a different axiom: *events have intensity, and that intensity is κ* . Not the price of time. The probability of resolution. Block rewards come from real trading activity, not from time passing. Validator staking earns from fees, not from inflationary dilution. Every instrument is priced from its own domain's real economic data with no discount rate anywhere in the calculation. A transaction implying a positive interest parameter is rejected at inclusion not because a smart contract refuses it but because the validator set's inclusion rules assert $\iota = 0$ — pinning the parameter to zero at consensus, with a proposer that violates it slashed.

Ethereum encodes $r > 0$ in its substrate. The κ -Chain encodes $\iota = 0$ in its substrate. That difference — one mathematical parameter, present in every block, every fee, every instrument, every validator reward — is the entire project. Not a product. Not a protocol. A different answer to the foundational question of financial mathematics.

13.3 What the competitive landscape revealed

The research on HAQQ, IsDB, QFC, ICS, and the Cosmos 2026 roadmap revealed something that was not obvious before it was investigated: the hardest engineering problems are already solved, and the ecosystem of Islamic financial institutions is ready for exactly what the κ -Chain provides.

IsDB holds a US patent on a consensus mechanism that ties validation authority to actual network usage. The institution that represents 57 Islamic Development Bank member countries, that has funded Islamic infrastructure

since 1975, is actively building blockchain consensus mechanisms. Qatar’s QFC has regulatory cover, institutional KYC infrastructure, and 33+ participating firms — and is running an Islamic finance blockchain POC on Hedera that has no pricing engine. It is compliance infrastructure looking for a monetary primitive to connect to. Bank Negara Malaysia and SAMA run dual monetary systems and need a market-determined Islamic rate to replace SOFR. Cosmos ICS provides 180+ validators for cold-start security. Cosmos PoA provides institutional validator onboarding without capital stake. IBC v2 connects the κ -Chain to every EVM chain and Solana without custom bridges.

The demand exists. The institutional readiness exists. The regulatory pathways exist. The engineering scaffolding exists. What has been missing — for fifty years — is the mathematical foundation for a riba-free monetary primitive. Papers 1–6 provide it.

13.4 The political economy conjunction

The Spring-Loaded Door computational political economy project (companion MASTER_FINDINGS.md) proved, across 16/16 binary classifications and 7/10 trajectory predictions, that 66% of states globally are in firing-classified mode — conditions where the L8 wealth-to-power converter is actively operating. The mechanism runs through L4 monetary plumbing: illiquidity is the bargaining position. The garment worker without income redundancy cannot refuse below-subsistence wages. The rice farmer who must sell at harvest cannot wait for a better price.

The conventional financial system’s L4 layer extracts a riba loading from every instrument that reaches these populations. Agricultural insurance in Bangladesh carries a 7.4% riba loading at $r = 8\%$ — extracted annually from subsistence farmers who have no awareness that the discount rate in their insurance contract constitutes a systematic transfer. Credit protection on OIC sovereign sukuk carries a present-value riba premium from discounting the future default loss. GCC central banks conducting Islamic monetary policy pay an implicit SOFR contamination in every sukuk they price.

These are not moral failings. They are structural features of a pricing system that was never designed to exclude interest. The riba is in the formula, not in anyone’s intention.

The κ -Chain removes the riba loading by removing the formula that generates it. The Bangladesh rice farmer’s takaful at $\pi^* = \kappa \cdot B$ — where κ is measured from historical loss frequencies ($\kappa_{MLE} = 0.071$ for Bangladesh; 0.071–0.147 across South and Southeast Asia) — costs what the drought risk actually costs. No additional charge for the dollar system’s existence. The garment worker’s micro-sukuk earns what κ implies, without a SOFR spread. The Malaysian sovereign prices its debt at the actual credit intensity of Malaysia, not at Malaysia’s credit intensity plus a premium for using Western financial infrastructure.

This is not charity. This is correct pricing. The riba loading is a systematic mispricing imposed by the structure of the conventional rate system on populations bearing real economic risk. The κ -Chain prices correctly by eliminating the source of the mispricing.

13.5 What this is

The 1.8 billion Muslims are not the market. They are the demonstration that the stakes are real. The market is every Islamic financial institution currently pricing sukuk against SOFR, every OIC central bank running a dual monetary system without a genuine riba-free benchmark, every agricultural takaful provider embedding a riba loading in premiums paid by subsistence farmers, every sovereign issuer paying a SOFR premium on debt that was structured to be halal.

The Baraka Protocol is Phase 1. It is already deployed. It proves the mechanism works with real money in a live market. The κ -Chain is Phase 3. It is the infrastructure that makes the mechanism the substrate — present in every block, every fee, every instrument, every validator reward — rather than a feature running on a conventional foundation.

The question the κ -Chain answers is not “how do we make Islamic finance compliant with blockchain?” The question is “what does a financial system look like when the interest rate is absent from its foundations, not as a constraint but as a mathematical consequence of pricing correctly?”

The answer: it looks like κ . A surface of real-world intensities, each domain measuring its own events, no committee, no discount rate, no price of time.

That is what fifty years of Islamic finance scholarship has been trying to build. That is what the mathematics now proves is possible. That is what this project — which started as a perpetual futures DEX and followed its own logic to its destination — set out to become.

References

- Ackerer, D., Hugonnier, J., & Jermann, U. (2025). Perpetual futures pricing. *Mathematical Finance*. DOI: 10.1111/mafi.70018. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4603820
- Ahmed, S., & Bhuyan, R. (2026a). The interest parameter in perpetual futures: Shariah analysis and empirical evidence from centralized and decentralized exchanges. SSRN 6322778. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6322778
- Ahmed, S., & Bhuyan, R. (2026b). From perpetual contracts to Islamic credit: The random stopping time equivalence and a riba-free foundation for sukuk, takaful, and credit protection. SSRN 6322858. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6322858
- Ahmed, S., & Bhuyan, R. (2026c). The κ -yield curve: Empirical estimation of the convergence intensity from sovereign sukuk spread data. SSRN 6322938. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6322938
- Ahmed, S., & Bhuyan, R. (2026d). Stochastic takaful pricing under CIR hazard dynamics: Application to agricultural insurance in South and Southeast Asia. SSRN 6323459. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6323459
- Ahmed, S., & Bhuyan, R. (2026e). The Islamic credit default swap: Riba-free credit protection with on-chain implementation. SSRN 6323519. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6323519
- Ahmed, S. (2026f). An integrated simulation framework for decentralised finance protocols: cadCAD, reinforcement learning, game theory, and mechanism design. SSRN 6323618. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6323618
- Ahmed, S., & Bhuyan, R. (2026g). The jurisprudential boundaries of an interest-free substrate: Riba, gharar, maysir, and qabd in cash-settled synthetic instruments. Working paper (companion research note).
- Ahmed, S., & Bhuyan, R. (2026h). Money as a mirror: Classical Islamic monetary ontology and the foundations of interest-free pricing. Working paper (companion paper).
- AAOIFI (2015). *Shariah Standards*. Accounting and Auditing Organization for Islamic Financial Institutions, Bahrain.
- Cox, J.C., Ingersoll, J.E., & Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- Duffie, D., & Singleton, K. (1999). Modeling term structures of defaultable bonds. *Review of Financial Studies*, 12(4), 687–720.
- El-Gamal, M.A. (2006). *Islamic Finance: Law, Economics, and Practice*. Cambridge University Press.
- Nelson, C.R., & Siegel, A.F. (1987). Parsimonious modeling of yield curves. *Journal of Business*, 60(4), 473–489.
- Svensson, L.E.O. (1994). Estimating and interpreting forward interest rates. NBER Working Paper 4871.

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